

Causal explanations of safety property violations in discrete event systems^{*}

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ABSTRACT

Autonomous decision-making increasingly relies on intricate embedded software, underscoring the need for automated explanations of system behaviour, especially in the presence of adverse events. The primary need is to swiftly pinpoint errors and instill confidence in the system's reliability. While traditional fault diagnosis discerns *what* happened in line with observations, this article delves deeper to elucidate *why* the observed outcome transpired, drawing inspiration from causal analysis techniques. Unlike traditional fault diagnosis, we consider a system to fail when a specific behavioural safety property is violated during its current operation. The formal explanations introduced in this context adhere to essential requirements that render them informative and pertinent. Furthermore, we explore the concept of contrastive explanations, which have demonstrated their ability to provide greater insight by comparing observed outcomes to expected ones. These explanations are causal in nature as they extract the pertinent aspects underlying safety property violations.

1. Introduction

As embedded software implements increasingly complex functionalities and takes over decision making even in critical situations, the ability to automatically construct explanations for the system behaviour is crucial during the whole software life cycle: at design time to quickly localise errors and gain confidence in the system, at runtime to understand the sources of failures, and post-mortem to help assigning legal liability for an accident.

The construction of formal explanations has been the subject of research work for a long time in different communities such as AI [1], molecular biology (explanation of how a product is brought about by a series of reactions) [2], and computer science (explanation of counterexamples from model checking) [3]. These communities developed, often independently, a set of techniques aiming at extracting relevant information from a system trace.

In this paper, we develop a theory of explanations for discrete event systems. We first formalise a set of expected requirements for explanations. In particular, explanations should be both *sound* with respect to the safety violation to be explained, that is, effectively

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pointing at a set of behaviours consistent with the observations and entailing the violation, and *complete*, meaning that they should cover all violating traces consistent with the observations. In addition, they should be *causal*, in the sense that they must only retain relevant information sufficient for entailing the violation. In addition, explanations are often more informative when they are given relative to some contrast case [4,5]. *Contrastive* explanations answer the question "Why did P happen rather than Q ?" and they provide more insight: instead of gathering all the causes of P , contrastiveness focuses the explanation towards the causes relative to the contrast case, i.e. what was expected. They are hence simpler.

We then discuss and put in perspective several notions of explanation in the light of these requirements and propose a new framework designed to overcome several issues. We address the construction of *formal explanations* for specific discrete event systems (DES), assuming that an explicit behavioural model is available. This model defines the entire context, i.e., all possible behaviours of the system. Our approach aims to derive causal explanations for safety violations directly from the model's interpretable components. The proposed explanations are thus inherently *interpretable*. Note that the techniques we employ differ fundamentally in nature and purpose from those used in eXplainable AI (XAI) [6,7] for data-driven black-box models. Whereas this XAI track focuses on elucidating statistical correlations learned from data, our framework seeks to extract and formalise the causal and structural justifications inherent in a well-defined model of system behaviour. They also differ from the explanations on symbolic AI systems that may take several forms like argumentative explanations or off-line justifications [8]. These are mostly rooted in logic, which is not the case of our framework. Further discussion of these distinctions is provided in Section 8 that reports related work.

Our approach lies at the crossroads between *model-based diagnosis* [9–11] and *causal analysis* [12–14]. The strength of a diagnostic process is usually to determine *what* happened that is consistent with the observations (consistency-based diagnosis) or to explain symptomatic observations with fault modes (abductive diagnosis based on logic entailment). Classically, model-based diagnosis of DES aims at detecting failures and isolating faulty event occurrences based on normal/abnormal behavioural models of the system used to check the consistency of an observable execution log [15–17]. Two important properties of a diagnoser are soundness (it returns only correct diagnoses) and completeness (it returns all the correct diagnoses) [18]. As stated above, the explanations must also possess these properties, defined in alignment with the diagnosers' properties. In this paper, we consider that a system fails as soon as a specific behavioural safety property is violated by the current run of the system and, to go beyond diagnosis, our objective is to explain *why* the violation has occurred based on the observed outcome. We pay attention to the insights from the social sciences and the findings of [4] stating not only that explanations should be contrastive but also that they should be "selected" to focus on a subset of causes.

The article [19] establishes the foundation stones of our work by proposing two types of explanations of the violation of a safety property P that focus on two different sets of potential causes: *subsequence explanations* and *choice explanations*. Subsequence explanations retain in the explanation a subsequence of events that are relevant to the violation of the safety property P . Although quite intuitive, we show that subsequence explanations, as defined in [19], have shortcomings when it comes to the desired properties of explanations. This is why this paper takes up and deepens the idea of choice explanations, which focus on explaining the outcome in terms of branching choices made by the system at hand.

Compared to [19], this article substantially enhances the framework. In particular, it enables the use of variables in addition to events through the formalism of implicit automata; it introduces a notion of semantics and formalises a set of expected properties based on these semantics; it casts subsequence and choice explanations in this framework; it significantly refines and extends our previous results on choice explanations; it proposes an efficient algorithm to compute the latter; it shows that choice explanations satisfy the expected properties; and it provides benchmarks showing that our approach is both effective and able to condense even complex executions to usually concise explanations.

Throughout the article, a system of two cars with adaptive cruise control is used to illustrate the introduced concepts, and the relevance of choice explanations is also demonstrated through a case study on a slice of the control software of the Therac-25 radiotherapy machine [20].

This article makes the following main contributions.

1. We establish a set of expected properties of explanations for dynamic systems, defined for the first time in terms of the *semantics* of an explanation.
2. We formalise a first type of explanations, referred to as *subsequence explanations*, that have been proposed by several communities to compress traces into shorter ones with the same effect. We discuss their usefulness and shortcomings in terms of the expected properties.
3. We propose a new class of explanations, called *choice explanations*, that focus on the *branching* behaviour as possible causes for the violation of a safety requirement in a game-based setting similar to [3], and study their satisfaction of the expected properties. To the best of our knowledge, our approach is the first construction of explanations for a given execution in dynamic systems that is *sound* and *complete* — i.e., an explanation correctly characterises only and all the system behaviors that violate the safety property —, and *precise*, i.e., each explanation is specific to the safety violation to be explained. We give some evidence that the generated explanations are also highly *concise*.
4. We propose a polynomial-time algorithm for the construction of choice explanations based on a pre-computation that is quadratic in the number of states of the model followed by the construction of the explanation for a given run that is linear in the number of states and in the length of the run.
5. We illustrate our approach on examples of safety-critical embedded systems. For all examples, the explanations have been computed with the prototype tool presented in Section 7 and used for extensive experimental evaluation (GitLab repository <https://gitlab.laas.fr/ypencole/ddexplain>).

The article is organised as follows. [Section 2](#) presents the formal framework that is used throughout the article and describes the problem statement. [Section 3](#) describes the case study of a two-car system and illustrates the concepts introduced in [Section 2](#). [Section 4](#) defines the properties that the explanations are expected to satisfy, discusses how we can define an explanation function to construct contrastive explanations, recalls the intuitive definition of an explanation as a subsequence of events, and shows why such definition is not satisfactory. [Section 5](#) presents the new definition of *choice explanation* and proposes an algorithm to compute choice explanations for the violation of a safety property, these being consistent with the observations of the system. The semantics of choice explanations are defined and their properties are proved. [Section 6](#) illustrates our approach on another example, a well-known software failure caused by *concurrency bugs*. [Section 7](#) shows on a set of benchmarks that choice explanations can be computed efficiently; even for complex models the obtained explanations usually encompass only a small set of contributory causes. [Section 8](#) presents related work and draws up a comparative analysis with respect to our proposal. Finally, [Section 9](#) summarises the contributions and outlines some perspectives for future work.

2. Framework

This section introduces the formal preliminaries and the problem statement. These preliminaries are illustrated with examples that are presented in the case studies in [Sections 3](#) and [6](#).

2.1. Formal preliminaries

Notations. Sets are denoted by upper case letters, given a set Y of sets we write $\bigcup Y$ for $\bigcup_{Z \in Y} Z$. Variables are denoted by lower case letters such as x_i , the domain of x_i is written $\mathbb{X}_i$ or $\text{dom}(x_i)$. Given a function $f : X \rightarrow Y$, its domain X is denoted by $\text{dom}(f)$. Let $X = \{x_1, \dots, x_n\}$ be a finite set of variables. Let $\mathbb{X} = \mathbb{X}_1 \times \dots \times \mathbb{X}_n$. For a valuation $(v_1, \dots, v_n) \in \mathbb{X}$ an $i \in [n]$ let $v(x_i) = v_i$ be the valuation of variable x_i . For sets X_1 and X_2 of variables with respective valuations v_1 and v_2 , v_1 and v_2 are *consistent* if $\forall x \in X_1 \cap X_2, v_1(x) = v_2(x)$. In that case we write $v_1 \cup v_2$ for the valuation of $X_1 \cup X_2$. For a finite set Y of variables, let $\text{pred}(Y)$ denote the set of predicates over Y . Given vectors $v_1 = \langle x_1, y_1 \rangle$ and $v_2 = \langle x_2, y_2 \rangle$, we write $v_1 < v_2$ when $x_1 \leq x_2 \wedge y_1 \leq y_2 \wedge v_1 \neq v_2$.

The framework that we use throughout this paper relies on DES represented by implicit automata.

Definition 1 (Implicit automaton). An implicit automaton is a tuple $\mathcal{A} = \langle \Sigma, X, \text{inv}, \delta, v^0, F \rangle$ where

- Σ is an alphabet of events;
- X is the set of variables;
- $\text{inv} \in \text{pred}(X)$ is an invariant;
- $\delta : \Sigma \rightarrow \text{pred}(X) \times \text{update}(X)$ where $\text{update}(X) = X \mapsto (\mathbb{X} \mapsto \bigcup \{\mathbb{X}_1, \dots, \mathbb{X}_n\})$ with $x_i \mapsto (\mathbb{X} \mapsto \mathbb{X}_i)$ is a partial function that defines an update;
- $v^0 \in \mathbb{X}_1 \times \dots \times \mathbb{X}_n$ is the *initial valuation*;
- $F \in \text{pred}(X)$ is the accepting predicate.

Implicit automata are a formalism that is rich enough to represent non trivial systems, through the concepts of events and variables while keeping transitions and discrete states implicit. As examples, [Section 3.1](#) describes the three implicit automata of the *two-car* system based on the events and the variables respectively presented in [Tables 1](#) and [2](#).

Given a valuation $v : X \mapsto \bigcup \{\mathbb{X}_1, \dots, \mathbb{X}_n\}$ and a partial update $f_i = (x_i \mapsto (\mathbb{X} \mapsto \mathbb{X}_i))$, we write $f_i(v)$ for the valuation v' obtained by applying f_i to the valuation v , that is, $v'(x) = f_i(x_i)(v)$ and $\forall x \in X \setminus \{x_i\} : v'(x) = v(x)$.

Definition 2 (Semantics of an implicit automaton). The semantic graph of an implicit automaton $\mathcal{A} = \langle \Sigma, X, \text{inv}, \delta, v^0, F \rangle$ is defined by the labeled transition system (LTS) $[\mathcal{A}] = \langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle$ where

- $\mathcal{V} = \{v \in \mathbb{X} \mid \text{inv}(v)\}$;
- $\rightarrow = \{(v, e, v') \in \mathcal{V} \times \Sigma \times \mathcal{V} \mid \delta(e) = \langle g, f \rangle \wedge g(v) \wedge v' = f(v)\}$;
- $\mathcal{V}^F = F \times \mathbb{X}$.

As usual we write $v \xrightarrow{e} v'$ for $(v, e, v') \in \rightarrow$. Given a set of nodes $\mathcal{V}' \subseteq \mathcal{V}$, let $\text{pre}_{\mathcal{A}}(\mathcal{V}') = \{v \in \mathcal{V} \mid \exists e \in \Sigma \exists v' \in \mathcal{V}' : v \xrightarrow{e} v'\}$ be the set of its immediate predecessor nodes. For a node v we also write $\text{pre}_{\mathcal{A}}(v)$, and v^* for the postset $\{v' \in \mathcal{V} \mid \exists e \in \Sigma : v \xrightarrow{e} v'\}$ when there is no ambiguity about the LTS.

By definition, all implicit automata are deterministic: for any event e and transitions $v \xrightarrow{e} v_1$ and $v \xrightarrow{e} v_2$ we have that $v_1 = v_2$ in their semantic graph. For instance, [Fig. 12](#) shows the semantic graph of the *Therac-25* case study.

In practice, implicit automata are rarely specified for an entire system. Typically, automata are defined for individual subsystems, which must then be composed (see [Sections 3.1](#) and [6](#) for examples). The composition of implicit automata, noted $\parallel$, as well as the notion of consistent updates referenced in the definition, are formalised in [Appendix A](#). Furthermore, the projection of a semantic state is defined with respect to a composed implicit automaton.

We distinguish the notion of *run* and *trace*.

Definition 3 (Run). Given an implicit automaton $\mathcal{A} = \langle \Sigma, X, \text{inv}, \delta, v^0, F \rangle$ with $[\mathcal{A}] = \langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle$, a *run* of $\mathcal{A}$ starting in state v_0 is a finite alternating sequence $r = v_0 \xrightarrow{e_0} v_1 \xrightarrow{e_1} \dots$ of states and labels of $[\mathcal{A}]$.

The length of a run $r = v_0 \xrightarrow{e_0} v_1 \xrightarrow{e_1} \dots \xrightarrow{e_{k-1}} v_k$ is $|r| = k$.

A run $r = v_0 \xrightarrow{e_0} v_1 \xrightarrow{e_1} \dots$ is *accepting* if $v_{|r|} \in \mathcal{V}^F$.

Let $\rho(\mathcal{A}, v_0)$ and $\rho^F(\mathcal{A}, v_0)$ denote the set of runs and the set of accepting runs of $\mathcal{A}$ starting in v_0 , respectively. We omit the second argument when it is v^0 , the initial state of $\mathcal{A}$.

A *trace* $tr \in \Sigma^*$ is a finite sequence of events. The concatenation of tr with some event a is written $tr \cdot a$. The length of tr is written $|tr|$ and it is equal to the number of occurrences of events.

Definition 4 (Trace). The *trace* of a run r , denoted $tr(r)$, is the sequence of events labeling the transitions of r . The trace of r is accepting if r is accepting.

The *language* $L(\mathcal{A}, v_0)$ (resp. *accepting language* $L^F(\mathcal{A}, v_0)$) of $\mathcal{A}$, is the set of traces (resp. accepting traces) from v_0 . Again we omit the second argument when it is the initial state of $\mathcal{A}$.

Given an implicit automaton $\mathcal{A}$ and a trace tr , let $\rho_{\mathcal{A}}(tr)$ denote the state reached by the (unique) run of $\mathcal{A}$ starting in the initial state of $\mathcal{A}$ and whose trace is tr .

Fig. 3 shows a run r_1 of the semantic graph of the *two-car* system. The associated trace $tr(r_1)$ is the sequence *fa.rcr.meas.fcr.ra.fail.meas*.

In the sequel, we restrict ourselves to models for which the state space reachable from the initial state is finite.

Definition 5 (Projection of a trace). Let Σ, Σ' be two sets of events such that $\Sigma' \subseteq \Sigma$. Let $\cdot_{\downarrow \Sigma'} : \Sigma^* \rightarrow \Sigma'^*$ be the projection defined by induction on traces:

$$\forall e \in \Sigma, e_{\downarrow \Sigma'} = \begin{cases} e \cap \Sigma' & \text{if } e \cap \Sigma' \neq \emptyset \\ \epsilon & \text{otherwise} \end{cases}$$

and $\forall t \in \Sigma^*, \forall e \in \Sigma : (t \cdot e)_{\downarrow \Sigma'} = t_{\downarrow \Sigma'} \cdot e_{\downarrow \Sigma'}$ where ϵ is the neutral element for the concatenation.

Definition 6 (Inevitability). Given an LTS $\mathcal{A} = \langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle$ and $\mathcal{V}_1, \mathcal{V}_2 \subseteq \mathcal{V}$, $\mathcal{V}_2$ is *inevitable* from $\mathcal{V}_1$ if for any $v \in \mathcal{V}_1$, any sufficiently long run in $\rho(\mathcal{A}, v)$ has some state in $\mathcal{V}_2$.

2.2. Problem statement

We address the problem of defining and computing different notions of explanation for the violation of a given behavioural safety property by a DES under partial observation. We consider that a behavioural model of the system is available and is represented as an implicit automaton $\mathcal{A} = \langle \Sigma, X, \text{inv}, \delta, v^0, F \rangle$. This automaton can be monolithic or can result from a multi-component model, in which case the system is modeled by a set of interacting implicit automata.

The system has a prefix-closed behavioural model. That means that every prefix of a trace of the system is also a trace of the system so $F = \text{true}$.

We assume the set Σ of events to be partitioned into *controllable events* Σ_c (e.g., events executed by the system) and *uncontrollable events* Σ_u (e.g., events executed by its environment). In a composition, we consider events shared by several components as controllable if they are controllable for all components; all other shared events are uncontrollable. In addition, we assume Σ to be partitioned into *observable events* Σ^o that are recorded and *non-observable events* Σ^{uo} that are not recorded, i.e. $\Sigma = \Sigma^o \uplus \Sigma^{uo}$. The two partitions are independent. In particular, there may be events that are controllable but not observable (e.g., due to memory constraints in the logging system), events that are uncontrollable but observable (e.g., relevant events received from the system environment), and events that are neither controllable nor observable.

We make the assumption that there are no unobservable cycles in $\mathcal{A}$, in other words we assume that the number of unobservable events that can be generated by the system between any two observable events is bounded. Any run of the system can be associated with a *log* that identifies a finite sequence of observable events. We consider that the log is an implicit automaton $\mathcal{L}$.

Definition 7 (Log). Let $\mathcal{A} = \langle \Sigma, X, \text{inv}, \delta, v^0, F \rangle$ be a system model and a run $r = v^0 \xrightarrow{e_0} v_1 \xrightarrow{e_1} \dots \xrightarrow{e_{k-1}} v_k \in \rho(\mathcal{A})$. The *log* of r with observable variables $X' \subseteq X$ is the implicit automaton $\mathcal{L}_{X'}(r) = \langle \Sigma^o, X' \cup \{q\}, \text{inv}', \delta', (v^0)', F' \rangle$ where

- q is a fresh variable ($q \notin X$) ranging over $\{0, \dots, \ell\}$, where ℓ is the number of observable events in r ;
- $\text{inv}' = \text{true}$;
- $(v^0)' = v^0_{\downarrow X'} \wedge q = 0$, where $v^0_{\downarrow X'}$ is the projection of v^0 on X' ;
- there exist functions $R : \{0, \dots, k\} \rightarrow \{0, \dots, \ell\}$, $g : \Sigma^o \times \{0, \dots, \ell\} \rightarrow \text{pred}(X' \cup \{q\})$, and $f : \Sigma^o \times \{0, \dots, \ell\} \rightarrow \text{update}(X' \cup \{q\})$ such that $R(0) = 0$, $R(k) = \ell$, $\forall e \in \Sigma^o$, $g(e, 0) = \text{false}$ and $f(e, 0) = \emptyset$, and $\forall i \in \{0, \dots, k-1\}$, if $R(i) = j$ for some j then
 - if $e_i \in \Sigma^o$ then $R(i+1) = j+1$, $g(e_i, j+1) = g(e_i, j) \vee q = j$, and $f(e_i, j+1) = f(e_i, j) \cup \{q = j \mapsto q := j+1\} \cup \{q = j \mapsto x := f_i(x) \mid x \in X' \cap \text{dom}(f_i) \wedge \delta(e_i) = \langle g_i, f_i \rangle\}$, and $\forall e \in \Sigma^o \setminus \{e_i\}$: $g(e, j+1) = g(e, j)$ and $f(e, j+1) = f(e, j)$
 - otherwise, $R(i+1) = j$
- and $\forall e \in \Sigma^o : \delta'(e) = \langle g(e, \ell), f(e, \ell) \rangle$;
- $F' = (q = \ell)$.

Note that the log is defined so as to enable, when composed with $\mathcal{A}$, exactly the set of prefixes of runs that produce the logged observations.

Example 1. Let $\mathcal{A} = \langle \Sigma, X, \text{inv}, \delta, \nu^0, F \rangle$ be a system model with $\Sigma = \Sigma^o \uplus \Sigma^{uo}$, $\Sigma^o = \{a\}$, and $\Sigma^{uo} = \{b\}$, $X = \{l, x, y\}$, $\text{inv} = \text{true}$, $\delta = \{a \mapsto (l = 0) \mapsto \{x := 1, y := 2, l := 1\}, b \mapsto (l = 1) \mapsto \{x := 3, y := 4, l := 0\}\}$, $\nu^0 = \langle 0, 0, 0 \rangle$, and $F = \text{true}$. Consider the run

$$r = \langle l = 0, x = 0, y = 0 \rangle \xrightarrow{a} \langle 1, 1, 2 \rangle \xrightarrow{b} \langle 0, 3, 4 \rangle \xrightarrow{a} \langle 1, 1, 2 \rangle$$

of $\mathcal{A}$. The log of r with the set of observable variables $X' = \{x\}$ is the implicit automaton $\mathcal{L}_{X'}(r) = \langle \Sigma^o, X' \cup \{q\}, \text{inv}', \delta', (\nu^0)', F' \rangle$ where

- $q = \{0, 1, 2\}$ and $\text{inv}' = \text{true}$;
- $R(i)$ is the number of observable events that have been seen before the $i - 1$ th state of the run r , that is $R(0) = 0, R(1) = 1, R(2) = 1, R(3) = 2$, as a is observable but not b , and
 $\delta'(a) = \{(q = 0) \mapsto \{x := 1, q := 1\}, (q = 1) \mapsto \{x := 1, q := 2\}\}$;
- $(\nu^0)' = \langle 0, 0 \rangle$, and $F' = (q = 2)$.

In other words, $\mathcal{L}_{X'}(r)$ allows exactly for the single run

$$\langle q = 0, x = 0 \rangle \xrightarrow{a} \langle 1, 1 \rangle \xrightarrow{a} \langle 2, 1 \rangle.$$

As the system is partially observable, a log $\mathcal{L}$ can be associated with several possible traces; these are the traces that are *consistent* with the log $\mathcal{L}$. Formally, the set of traces consistent with the log $\mathcal{L}$ is given by:

$$\text{tr}(\mathcal{L}) = \{t \in \Sigma^* \mid t|_{\Sigma_o} = \mathcal{L}\} \cap L(\mathcal{A})$$

On the other hand, we also assume that the set X of variables includes a subset of *observable variables* $X' \subseteq X$, while the others remain unobserved. The valuation of an observable variable is observed each time an observable event makes an assignment to it.

Throughout this article we focus on explaining the violation of a safety property. A safety property P characterises a set of runs of a system such that whenever a run r does not satisfy P , then no run having r as a prefix satisfies P . For instance, once the safety property “state ν must never be entered” is violated, it remains violated forever. Following current practice, we use *observers* in the form of (implicit) automata to represent safety properties. This allows us to directly use the observer to track when the safety property has been violated.

Definition 8 (Observer). Given implicit automata $\mathcal{A} = \langle \Sigma, X, \text{inv}, \delta, \nu^0, F \rangle$ and $\mathcal{O} = \langle \Sigma_{\mathcal{O}}, X_{\mathcal{O}}, \text{inv}_{\mathcal{O}}, \delta_{\mathcal{O}}, \nu_{\mathcal{O}}^0, ko \rangle$, we say that $\mathcal{O}$ is an *observer* for $\mathcal{A}$ if

- the accepting predicate ko is inductively invariant, that is, the post-condition of all transitions enabled in ko satisfies ko ;
- $\mathcal{O}$ is receptive: $\forall v \forall e \in \Sigma_{\mathcal{O}} \exists g, f : \delta_{\mathcal{O}}(e) = \langle g, f \rangle \wedge g(v) \wedge \text{inv}_{\mathcal{O}}(f(v))$; and
- $\Sigma_{\mathcal{O}} \subseteq \Sigma$ and $\forall \langle e, \langle g, f \rangle \rangle \in \delta_{\mathcal{O}} \forall x \in X : x \notin \text{dom}(f)$.

That is, an observer may observe a subset $\Sigma_{\mathcal{O}} \subseteq \Sigma$ of the events and a subset $X_{\mathcal{O}} \cap X$ of the variables of the observed model $\mathcal{A}$. It may also have “private” variables $X_{\mathcal{O}} \setminus X$, for instance, to count events. The observer does not update variables that are shared with $\mathcal{A}$. The observer entering a state that satisfies ko amounts to the violation of the safety property. Section 3.4 details the observer used in the two-car system.

Notice that a log and an observer do not necessarily observe the same events and variables. Let Σ^o , $\Sigma_{\mathcal{O}}$, X' , and $X_{\mathcal{O}}$ denote the observable system events, the events of an observer, the variables observable by a log, and the variables of an observer, respectively. We distinguish the following particular cases:

- $\exists e \in \Sigma^o \cap \Sigma_{\mathcal{O}}$: event e is observable by the log and relevant for the observer.
- $\exists e \in \Sigma_{\mathcal{O}} \setminus \Sigma^o$: event e is not observable by the log but relevant for the observer. This case is taken into account since $\mathcal{A} \parallel \mathcal{O} \parallel \mathcal{L}$ may produce all runs of $\mathcal{A}$ that are consistent with the log $\mathcal{L}$.
- $\exists e \in \Sigma^o \setminus \Sigma_{\mathcal{O}}$: event e is logged but not pertinent for the observer.
- $\exists x \in X' \cap X_{\mathcal{O}}$: variable x is observable by the log, and relevant for the observer. If x is updated by observable events, then the assignment is immediately visible in the log. In contrast, if x is updated by an event not in Σ^o , then the assignment is not observed. Again, this case is taken into account since $\mathcal{A} \parallel \mathcal{O} \parallel \mathcal{L}$ may produce all runs of $\mathcal{A}$ that are consistent with the log $\mathcal{L}$, including those where x is updated by an unobservable event.
- $\exists x \in X_{\mathcal{O}} \setminus X'$: variable x is not observable by the log but relevant for the observer. This case is taken into account since $\mathcal{A} \parallel \mathcal{O} \parallel \mathcal{L}$ may produce all runs of $\mathcal{A}$ that are consistent with the log $\mathcal{L}$.
- $\exists x \in X' \setminus X_{\mathcal{O}}$: variable x is logged but not pertinent for the observer.

From a practical perspective, it is important that the logged information be consistent with the model and the expected safety property. If information required by the model is missing, spurious runs may match the log, leading to explanations not relevant to the actual run. Likewise, missing information about how the safety property was violated (i.e., which path in the observer was taken) may result in part of the explanations not being pertinent for the actual run. Conversely, logging more information than is necessary to uniquely identify the actual run of $\mathcal{A} \parallel \mathcal{O}_p$ does not yield more informative explanations.

As the mere language accepted by an observer in isolation does not reflect its state constraints, we define below the safety property it models when composed with a model to be observed.

Given an observer $\mathcal{O} = \langle \Sigma_{\mathcal{O}}, X_{\mathcal{O}}, \text{inv}_{\mathcal{O}}, \delta_{\mathcal{O}}, v_{\mathcal{O}}^0, ko \rangle$ for a behavioural model $\mathcal{A}$ with $[\mathcal{A} \parallel \mathcal{O}] = \langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle$, let $ok_{\mathcal{A}}(\mathcal{O}) = \{v \in \mathcal{V} \mid \neg ko(v_{\mathcal{O}})\}$ be the set of states in which $\mathcal{O}$ is not in its accepting state.

Considering $\mathcal{V}' = ok_{\mathcal{A}}(\mathcal{O})$, the set of words reaching exactly the set of safe states with respect to $\mathcal{V}'$ in $[\mathcal{A} \parallel \mathcal{O}]$ can be defined as follows.

Definition 9 (Induced safe language). Given a behavioural model $\mathcal{A}$ over Σ , an observer $\mathcal{O}$ with $[\mathcal{A} \parallel \mathcal{O}] = \langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle$, and an initial state $v_0 \in \mathcal{V}$, the safe language induced by $\mathcal{O}$ on $\mathcal{A}$ from v_0 with respect to a set of uncontrollable events $U \subseteq \Sigma$, written $\text{safeLang}_{\mathcal{A}, U}(\mathcal{O}, v_0)$, is the greatest language $X \subseteq L(\mathcal{A}, (v_0)_{\downarrow \mathcal{A}}) \setminus L^F(\mathcal{A} \parallel \mathcal{O}, v_0)$ such that $\forall w \in X$:

- $\exists e \in \Sigma : w \cdot e \in X$, and
- $\forall e \in U : (w \cdot e \in L(\mathcal{A}, (v_0)_{\downarrow \mathcal{A}})) \implies w \cdot e \in X$.

We omit the argument v_0 when $v_0 = v^0$, and the parameter U when we denote the safety language with respect to all the uncontrollable events of $\mathcal{A}$, i.e. $U = \Sigma^u$.

Note that we rely on the determinism of the automata to ensure that the induced safe language is exactly the language that keeps the product $[\mathcal{A} \parallel \mathcal{O}]$ outside of the state space in which the observer is in its *ko* states.

Considering an observer $\mathcal{O}_P$ that observes the safety property P , we can now formally represent as an implicit automaton:

- the set of runs of the system that violate the safety property P : $\mathcal{A} \parallel \mathcal{O}_P$;
- the set of runs of the system that are consistent with a given log $\mathcal{L}$: $\mathcal{A} \parallel \mathcal{L}$;
- the set of runs of the system that violate the safety property P , and that are consistent with a given log $\mathcal{L}$: $\mathcal{A} \parallel \mathcal{O}_P \parallel \mathcal{L}$.

Note that, synchronisation being an associative operation, $\mathcal{A} \parallel \mathcal{O}_P \parallel \mathcal{L}$ is perfectly defined, and $\mathcal{A} \parallel \mathcal{O}_P \parallel \mathcal{L} = (\mathcal{A} \parallel \mathcal{O}_P) \parallel \mathcal{L} = \mathcal{A} \parallel (\mathcal{O}_P \parallel \mathcal{L})$. A state of $\mathcal{A} \parallel \mathcal{O}_P \parallel \mathcal{L}$ is a tuple $\langle v_a, v_p, v_l \rangle$ where v_a, v_p , and v_l are the respective states in $\mathcal{A}, \mathcal{O}_P$, and $\mathcal{L}$.

Given a system model $\mathcal{A}$, a safety property P , and a log $\mathcal{L}$ violating P , we aim at constructing a causal explanation for this violation. More precisely, our goal is to identify what in the controllable behaviour is consistent with the log and can be blamed for the violation, and to which extent the uncontrollable events come into play.

3. Case study

An example inspired from the autonomous car domain will be used all along the paper to illustrate our proposal.

The system, named the *two-car system*, considers a car equipped with a cruise controller and a sensor, named the *rear car* following another car named the *front car*. The sensor measures the distance of the rear car with respect to the front car as shown in Fig. 1.

The positions of the front car fcx and of the rear car rcx evolve according to their respective speeds $fcs \in [0, V_{\max}]$ and $rfs \in [0, V_{\max}]$. The updates after one time unit $\delta t = 1$ are given by:

$$fcx \leftarrow fcx + \delta t \times fcs, \quad (1)$$

$$rcx \leftarrow rcx + \delta t \times rcs, \quad (2)$$

We assume that the front car is free to choose its speed in $\{0, \dots, V_{\max}\}$ and that the rear car cruise controller is designed to avoid collision in any situation, i.e. to maintain some distance between the two cars.

At any time, the distance between the two cars sd is given by:

$$sd = fcx - rcx. \quad (3)$$

Using (1) in (3), we can derive the following update of the distance after one time unit:

$$\begin{cases} sd \leftarrow sd - rcs + fcs & \text{if } sd + fcs \geq rcs \\ sd \leftarrow 0 & \text{otherwise} \end{cases} \quad (4)$$

We also assume that the sensor may fail and, if it fails, it reports a measured distance smd that is twice the real distance sd . The sensor mode $sfm \in \{0, 1\}$, where 0 indicates the fault mode and 1 indicates the normal mode, hence affects the behaviour of the cruise controller.

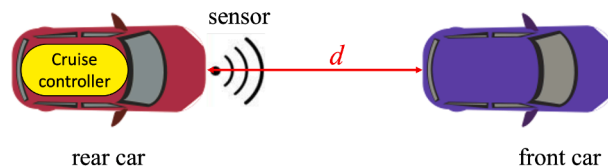


Fig. 1. Two-car system.

Table 1
Events of the model.

Name	Description
<i>ra</i>	rear car acceleration
<i>rbr</i>	rear car break
<i>rcr</i>	rear car cruise
<i>fa</i>	front car acceleration
<i>fbr</i>	front car break
<i>fcr</i>	front car cruise
<i>fail</i>	sensor fault event
<i>meas</i>	sensor distance measurement

Table 2
Variables of the model ($V_{\max} = 2$, $smdsat = 10$).

Name	Description	Domain	Components
<i>rcs</i>	rear car speed	$\mathbb{X}_{rcs} = \{0, \dots, V_{\max}\}$	CC, SR
<i>fcs</i>	front car speed	$\mathbb{X}_{fcs} = \{0, \dots, V_{\max}\}$	FC, SR
<i>sd</i>	real distance between cars	$\mathbb{X}_{sd} \geq 0$	SR
<i>smd</i>	measured distance between cars	$\mathbb{X}_{smd} = \{0, \dots, smdsat\}$	CC, SR
<i>sfm</i>	sensor fault mode	$\mathbb{X}_{sfm} = \{0, 1\}$	SR
<i>cm</i>	communication mode	$\mathbb{X}_{cm} = \{0, 1, 2\}$	CC, FC, SR

The question to be answered is: what kind of *explanations* can we provide about the violation of the safety property P that ensures $sd > 0$ at any time, i.e. no collision of the rear car into the front car, based on the log $\mathcal{L}$, the model $\mathcal{A}$ and the observer $\mathcal{O}_P$ as defined below?

The two-car system is represented as three interacting components modelled by three implicit automata:

1. An automaton *Front Car* (FC for short) that models the speed of the front car (*fcs*). The front car can accelerate (*fa*), break (*fbr*) or cruise (*fcr*).
2. An automaton *Sensor* (SR for short) that models the sensor that aims at measuring (measure *smd*) the distance *sd* between the controlled car and the front car. The sensor may fail (event *fail*), if it fails it provides as a measured distance (*smd*) a value that is twice the real distance (*sd*) which affects the behaviour of the cruise controller. The sensor saturates as soon as *smd* reaches the value *smdsat*.
3. An automaton *Cruise Controller* (CC for short) that models the cruise controller of the car behind the front car. The controller can accelerate (*ra*), break (*rbr*) or cruise (*rcr*), depending on a measured distance (*smd*) provided by a distance sensor. When the sensor saturates, the cruise controller systematically applies *ra* so that the rear car reaches max speed and keeps it. If the distance between the two cars *sd* reaches 0, there is a collision and no further action is triggered.

Fig. 2 presents the structure of the model of the system. The set of events of this model is detailed in Table 1. The set of variables, their domains, and the components using them are detailed in Table 2.

Events *ra*, *rbr*, *rcr* occur in the cruise controller. Events *fa*, *fbr*, *fcr* occur in the front car. The events *fail* and *meas* occur in the sensor.

The rear car and front car accelerations *ra* and *fa* increase the rear car and front car speed by 1 unit. The rear car and front car break *rbr* and *fbr* decrease the rear car and front car speed by 1 unit. The cruise *rcr* and *fcr* actions maintain the speeds of the rear car and front car, respectively. The event *fail* indicates that a fault occurs in the sensor and the *sfm* fault mode variable is immediately updated. The event *meas* is the measurement event in the sensor that updates the measured distance between the two cars *smd*. Next measure *smd* of the sensor will double the real distance *sd*. In any case, the sensor saturates as soon as *smd* reaches the value *smdsat*.

For every action of the front car (*fa*, *fbr*, *fcr*), the cruise controller responds with an action to apply to the rear car (i.e. *ra*, *rbr*, *rcr*) then gives control to the sensor. The communication protocol between the front car, the cruise controller, and the sensor is orchestrated by the shared variable *cm*. After every action of the front car (communication mode *cm* = 0), the cruise controller applies an action on the rear car (communication mode *cm* = 1) and the sensor updates the current distance based on these two actions (communication mode *cm* = 2).

Variables *rcs* and *smd* are shared by the automata of the cruise controller and the sensor. The rear car speed *rcs* is indeed required by the sensor to update the distance measure *smd* that is used in turn by the cruise controller to update the car speed *rcs* by choosing one of the events *ra*, *rbr*, *rcr*. Variable *fcs* is shared between the sensor and the front car automata, hence the update of the measured distance *smd* by the sensor also requires the front car speed *fcs*. Variables *sd*, *sfm* are internal to the sensor automaton.

The initial situation of the system is $v^0 = ((v^0)_{SR}, (v^0)_{FC}, (v^0)_{CC}) = (cm = 0, fcs = 1, rcs = 0, sd = 3, sfm = 1, smd = 3)$: the car is stopped at distance 3 of the front car whose speed is 1, with a correctly working sensor.

The behaviour of the *two-car* system is obtained as $\mathcal{A} = FC \parallel CC \parallel SR$ by composing the implicit automata of the front car FC, the cruise controller CC, and the sensor SR models that are provided below.

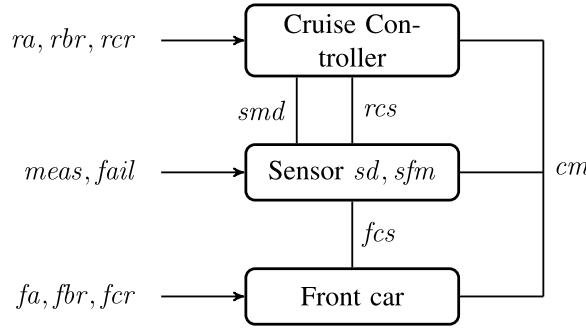


Fig. 2. Two-car system: structural model.

3.1. Models

This section presents the implicit automata of the *two cars* system (see Definition 1).

3.1.1. Sensor model SR

$$SR = \langle \Sigma_{SR}, X_{SR}, inv_{SR}, \delta_{SR}, (v^0)_{SR}, F_{SR} \rangle$$

- $\Sigma_{SR} = \{meas, fail\}$
- $X_{SR} = \{sd, rcs, fcs, cm, smd, sfm\}$
- $inv_{SR} = smd \leq smdsat$
- Transitions:
 - $\delta_{SR}(fail) = (sfm = 1, \{sfm \leftarrow 0\})$
 - $\delta_{SR}(meas) = (cm = 2, \{sd \leftarrow \max(sd - rcs + fcs, 0), smd \leftarrow \min[(2 - sfm)sd, smdsat], cm \leftarrow 0\})$
- $(v^0)_{SR} = (sd = 3, sfm = 1, smd = 3, cm = 0, rcs = 0, fcs = 1)$
- $F_{SR} = inv$

3.1.2. Front car model FC

$$FC = \langle \Sigma_{FC}, X_{FC}, inv_{FC}, \delta_{FC}, (v^0)_{FC}, F_{FC} \rangle$$

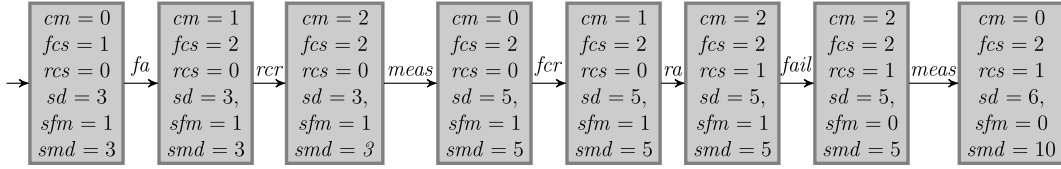
- $\Sigma_{FC} = \{fa, fbr, fcr\}$
- $X_{FC} = \{cm, fcs\}$
- $inv_{FC} = fcs \in [0, V_{\max}]$
- Transitions:
 - $\delta_{FC}(fa) = (cm = 0, \{cm \leftarrow 1, fcs \leftarrow fcs + 1\})$;
 - $\delta_{FC}(fbr) = (cm = 0, \{cm \leftarrow 1, fcs \leftarrow fcs - 1\})$;
 - $\delta_{FC}(fcr) = (cm = 0, \{cm \leftarrow 1\})$
- $(v^0)_{FC} = (cm = 0, fcs = 1)$
- $F_{FC} = inv$

3.1.3. Cruise controller model CC

$$CC = \langle \Sigma_{CC}, X_{CC}, inv_{CC}, \delta_{CC}, (v^0)_{CC}, F_{CC} \rangle$$

- $\Sigma_{CC} = \{ra, rbr, rcr\}$
- $X_{CC} = \{cm, smd, rcs\}$
- $inv_{CC} = rcs \in [0, V_{\max}]$
- Transitions:
 - $\delta_{CC}(ra) = (cm = 1 \wedge smd > 2(rcs + 1), \{cm \leftarrow 2, rcs \leftarrow rcs + 1\})$;
 - $\delta_{CC}(rbr) = (cm = 1 \wedge rcs \geq 1 \wedge 0 < smd < smdsat, \{cm \leftarrow 2, rcs \leftarrow rcs - 1\})$;
 - $\delta_{CC}(rcr) = (cm = 1 \wedge smd > 2rcs \wedge (smd < smdsat \vee (smd = smdsat \wedge rcs = V_{\max})), \{cm \leftarrow 2\})$
- $(v^0)_{CC} = (cm = 0, rcs = 0, smd = 2)$
- $F_{CC} = inv$

Note that, before applying an action that decreases the distance between the two cars, the cruise controller anticipates the braking distance (brd , which is equal to rcs , in the transition guards). Therefore, we must have $smd_{updated} > brd = rcs_{updated}$ and, as we have $smd_{updated} = (smd - rcs_{updated})$, it is required that $(smd - rcs_{updated}) > rcs_{updated}$, implying $smd > 2rcs_{updated}$. Consequently, since $rcs_{updated} = rcs + 1$ for the acceleration action ra , the guard of the transition $\delta_{CC}(ra)$ is $smd > 2(rcs + 1)$. For the cruise action rcr ,

Fig. 3. A run r_1 of $[A] = [FC||CC||SR]$.

$r_{cs_updated} = r_{cs}$, so the guard of the transition $\delta_{CC}(rcr)$ is $smd > 2r_{cs}$. No guard is required for the breaking action rbr because it does not harm the breaking distance. The above conditions in the transition guards also ensure that no further action is triggered if $sd = 0$ as it implies $smd = 0$.

3.2. Semantic graph of the model

The semantic graph of the model $A = FC||CC||SR$ is $[A] = [FC||CC||SR]$ (cf. Definition 2).

Fig. 3 presents a run of the system (see Definition 3). Starting from the initial state, the front car first accelerates (fa), which increases its speed at $fcs = 2$ while the rear car is still stopped ($r_{cs} = 0$). After the first sensor measurement, the real distance between the front car and the rear car is updated to $sd = 5 = \max(sd - r_{cs} + fcs, 0) = \max(3 - 0 + 2, 0)$ and the measured distance to $smd = 5 = \min[(2 - sfm)sd, smdsat] = \min[(2 - 1)5, 10] = 5$. After a cruise of the front car (fcr), the rear car accelerates (ra) and its speed goes to $r_{cs} = 1$ then the failure occurs on the sensor ($fail$). After the second measurement, the rear car believes that the distance is $smd = smdsat = 10$ (the sensor is saturated) while the actual distance is $sd = 6 = 5 - 1 + 2$. The trace $tr(r_1)$ is the sequence $fa.rcr.meas.fcr.ra.fail.meas$ (see Definition 4).

3.3. Log

Consider the run in Fig. 3 and assume that the observable events are $\Sigma^o = \{rcr, ra, rbr, meas\}$, i.e. the actions that are performed by the rear car. Also consider that the set of observable variables is $X' = \{r_{cs}, smd\}$, including the measured distance returned by the sensor and the speed of the rear car.

In run r_1 , there are four occurrences of an observable event, hence from Definition 7 $q \in \{0, 1, 2, 3, 4\}$: event $meas$ occurs twice (for $q = 1$ and $q = 3$) and both events rcr, ra occur once (for $q = 0$ and $q = 2$, respectively). Given that the valuation of an observable variable is observed only when an observable event makes an assignment to it, the observable variable smd is observed when the measurement event $meas$ occurs and r_{cs} when rcr or ra occur.

It follows from Definition 7 that $\mathcal{L}_{X'}(r_1) = \langle \Sigma^o, X' \cup \{q\}, inv', \delta', (v^0)', F' \rangle$ is such that:

- $q \in \{0, 1, 2, 3, 4\}$
- $inv' = true$
- Transitions:
 - $\delta'(meas) = \left(q \in \{1, 3\}, \left\{ \begin{array}{l} (q \leftarrow 2, smd \leftarrow 5) \text{ if } (q = 1) \\ (q \leftarrow 4, smd \leftarrow 10) \text{ if } (q = 3) \end{array} \right\} \right)$
 - $\delta'(rcr) = (q = 0, \{ (q \leftarrow 1, r_{cs} \leftarrow 0) \text{ if } q = 0 \})$
 - $\delta'(ra) = (q = 2, \{ (q \leftarrow 3, r_{cs} \leftarrow 1) \text{ if } q = 2 \})$
 - $\delta'(rbr) = (false, \emptyset)$
- $(v^0)' = (smd = 3, q = 0)$
- $F' = (q = 4)$

3.4. Observer

In this example, we consider the safety property P that ensures that at any time, the distance between the rear car and the front car is greater than 0 ($sd > 0$), i.e., a run violates P if it contains at least a state such that $sd = 0$, which amounts to a collision. To track the satisfaction or violation of P , the observer $\mathcal{O}_P = \langle \Sigma_{\mathcal{O}_P}, X_{\mathcal{O}_P} \cup \{q\}, inv_{\mathcal{O}_P}, \delta_{\mathcal{O}_P}, v_{\mathcal{O}_P}^0, ko_{\mathcal{O}_P} \rangle$ is designed herebelow according to Definition 8:

- $\Sigma_{\mathcal{O}_P} = \{meas\}$
- $X_{\mathcal{O}_P} = \{sd, fcs, rcs, pko\}$
- $inv_{\mathcal{O}_P} = true$
- Transitions:
 - $\delta_{\mathcal{O}_P}(meas) = (true, \{pko \leftarrow true, \text{ if } (\neg pko \wedge sd + rcs - fcs = 0)\})$
- $v_{\mathcal{O}_P}^0 = (pko = false, fcs = 1, rcs = 0, sd = 3)$
- $ko_{\mathcal{O}_P} = pko$

In this example, the safety property P is violated as soon as the system reaches a state with $sd = 0$. Therefore, the observer $\mathcal{O}_P$ only needs to keep track of the events that update sd , which are only *meas* events. To implement the accepting predicate $ko_{\mathcal{O}_P}$, a Boolean variable pko is introduced in the observer and initially set to *false*. As long as the current measure update *meas* leads to a state where $sd > 0$, pko remains *false*. When the current distance sd is about to be updated with $(sd + rcs - fcs)$ to become 0, pko is set to *true* and will remain true forever. For the observer to be receptive, i.e., always ready to accept any input event, the precondition of $\delta_{\mathcal{O}_P}(\text{meas})$ is always true.

4. Explanations and their expected properties

4.1. Properties of explanations

To reason formally about explanations, we must first provide them with a well-defined semantics. Intuitively, the semantics of an explanation is the set of traces that it explains. Furthermore, to make explanations meaningful and comparable, we formalise a set of key properties they should satisfy. These properties are parametrised with the semantics of explanations.

Given a model $\mathcal{A}$, a trace tr , and an observer $\mathcal{O}_P$, consider an *explanation function* $\mathcal{E}$ such that $\mathcal{E}(\mathcal{A}, \mathcal{O}_P, v_0, tr)$ is a set of possible explanations for the violation of P in tr starting in v_0 . We omit the latter argument v_0 when it is the initial state of $\mathcal{A}$.

A log $\mathcal{L}$ is associated with a set of traces, allowing the definition of explanations for $\mathcal{L}$ as follows:

$$\mathcal{E}(\mathcal{A}, \mathcal{O}_P, v_0, \mathcal{L}) = \bigcup_{tr \in L^F(\mathcal{A} \parallel \mathcal{L} \parallel \mathcal{O}_P, v_0)} \mathcal{E}(\mathcal{A}, \mathcal{O}_P, v_0, tr), \quad (5)$$

An explanation $\varepsilon \in \mathcal{E}(\mathcal{A}, \mathcal{O}_P, v_0, tr)$ uniquely defines a set of runs of $\mathcal{A}$ that are said to be covered by ε . We refer to the corresponding set of traces as $sem_{\mathcal{A}}(\varepsilon)$, representing the semantics of ε . Hence, $sem_{\mathcal{A}}(\varepsilon) \subseteq L(\mathcal{A})$. Note that, in our framework, there is a one to one correspondence between traces of $L(\mathcal{A})$ and runs of $\mathcal{A}$, then $sem_{\mathcal{A}}(\varepsilon)$ also refers unequivocally to a set of runs of $\mathcal{A}$.

To assess the quality of explanation functions, we define formal requirements with respect to the safety property and to the trace, expressed through specific properties.

Property 1 (Soundness). An explanation $\varepsilon \in \mathcal{E}(\mathcal{A}, \mathcal{O}_P, v_0, tr)$ is sound with respect to the safety property observed by $\mathcal{O}_P$ if

$$sem_{\mathcal{A}}(\varepsilon) \cap safeLang_{\mathcal{A}, \emptyset}(\mathcal{O}_P, v_0) = \emptyset$$

An explanation function $\mathcal{E}$ is sound with respect to $\mathcal{O}_P$ if for any $\varepsilon \in \mathcal{E}(\mathcal{A}, \mathcal{O}_P, v_0, tr)$, ε is sound with respect to the safety property observed by $\mathcal{O}_P$.

Intuitively, ε is sound if the extensions of all traces in the semantics of ε eventually violate the observed safety property. In other words, a sound explanation represents a set of traces for which the safety violation is unavoidable.

Property 2 (Completeness). An explanation function $\mathcal{E}$ is complete if for any $tr \in L^F(\mathcal{A} \parallel \mathcal{O}_P, v_0)$:

$$\forall \varepsilon \in \mathcal{E}(\mathcal{A}, \mathcal{O}_P, v_0, tr) : tr \in sem_{\mathcal{A}}(\varepsilon)$$

Completeness requires the trace tr violating the safety property observed by $\mathcal{O}_P$ to be covered by the explanation.

Property 3 (Precision). Consider a model $\mathcal{A}$, two observers $\mathcal{O}_{P_1}, \mathcal{O}_{P_2}$ with disjoint sets of private variables, and traces tr_i of $\mathcal{A} \parallel \mathcal{O}_{P_i}$ starting in $v_{0,i} = \langle v_{\mathcal{A},0}, v_{P_i,0} \rangle$, $i = 1, 2$. An explanation function $\mathcal{E}$ is explanation-precise with respect to $\mathcal{O}_{P_1}$ and $\mathcal{O}_{P_2}$ if

$$L^F(\mathcal{A} \parallel \mathcal{O}_{P_1} \parallel \mathcal{O}_{P_2}, v_0) = \emptyset \implies \mathcal{E}(\mathcal{A}, \mathcal{O}_{P_1}, v_{0,1}, tr_1) \cap \mathcal{E}(\mathcal{A}, \mathcal{O}_{P_2}, v_{0,2}, tr_2) = \emptyset \quad (6)$$

where $v_0 = \langle v_{\mathcal{A},0}, v_{P_1,0}, v_{P_2,0} \rangle$. It is semantics-precise if for any $\varepsilon_i \in \mathcal{E}(\mathcal{A}, \mathcal{O}_{P_i}, v_{0,i}, tr_i)$, $i = 1, 2$,

$$L^F(\mathcal{A} \parallel \mathcal{O}_{P_1} \parallel \mathcal{O}_{P_2}, v_0) = \emptyset \implies sem_{\mathcal{A}}(\varepsilon_1) \cap sem_{\mathcal{A}}(\varepsilon_2) = \emptyset \quad (7)$$

Intuitively, explanation-precision and semantics-precision mean, respectively, that two disjoint safety violations have disjoint sets of explanations, and any two explanations for the two safety violations have disjoint semantics.

Completeness on the one hand, and soundness and precision on the other hand are antinomic, in the sense that completeness defines a lower bound on the semantics of an explanation, whereas soundness and semantic precision define an upper bound. Together they ensure explanations to be specific to the trace and the safety violation to be explained.

Proposition 1. Consider a model $\mathcal{A}$, two observers $\mathcal{O}_{P_1}, \mathcal{O}_{P_2}$ with disjoint sets of private variables, and traces tr_1 and tr_2 of $\mathcal{A}$ violating P_1 and P_2 respectively. Any explanation function $\mathcal{E}$ that is sound with respect to $\mathcal{O}_{P_1}$ and $\mathcal{O}_{P_2}$ is semantics-precise. If in addition it is complete, then it is explanation-precise.

Proof. By soundness, $\forall \varepsilon_i \in \mathcal{E}(\mathcal{A}, \mathcal{O}_{P_i}, v_{0,i}, tr_i) : sem_{\mathcal{A}}(\varepsilon_i) \cap safeLang_{\mathcal{A}, \emptyset}(\mathcal{O}_{P_i}) = \emptyset$, $i = 1, 2$. Hence, from the state reached by any word $w \in sem_{\mathcal{A}}(\varepsilon_i)$, the violation of the safety property observed by $\mathcal{O}_{P_i}$ is inevitable. Eq. (7) follows.

Now suppose that $\varepsilon \in \mathcal{E}(\mathcal{A}, \mathcal{O}_{P_1}, v_{0,1}, tr_1) \cap \mathcal{E}(\mathcal{A}, \mathcal{O}_{P_2}, v_{0,2}, tr_2)$. By Eq. (7), it follows that $sem_{\mathcal{A}}(\varepsilon) = \emptyset$, which is in contradiction with completeness. Hence $\mathcal{E}$ is explanation-precise. $\square$

Property 4 (Monotony of explanations). Given a model $\mathcal{A}$, an observer $\mathcal{O}_P$, a couple $(\mathcal{E}, sem_{\mathcal{A}})$ is monotonic if for all v_0 and $tr_i \in L^F(\mathcal{A} \parallel \mathcal{O}_P, v_0)$ and $\varepsilon_i \in \mathcal{E}(\mathcal{A}, \mathcal{O}_P, v_0, tr_i)$, $i \in \{1, 2\}$, we have

$$tr_2 \in sem_{\mathcal{A}}(\varepsilon_1) \implies sem_{\mathcal{A}}(\varepsilon_2) \subseteq sem_{\mathcal{A}}(\varepsilon_1)$$

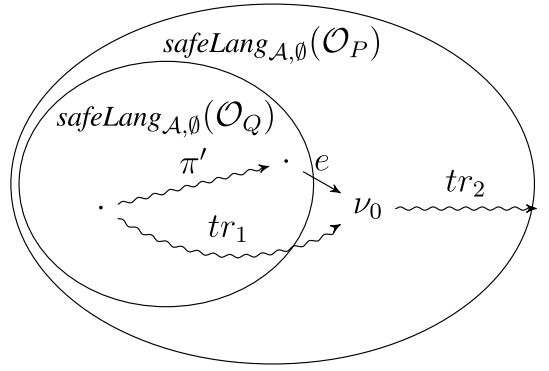


Fig. 4. Illustration of Definition 10.

Monotony implies that if an explanation of tr_1 covers tr_2 , then the explanation of tr_2 is smaller than the explanation of tr_1 in terms of semantics inclusion.

In addition to the formal requirements stated above, we argue that the semantics of an explanation should not depend on $\mathcal{O}_P$. In other words, the semantics of an explanation should only depend on the information retained in the explanation.

Remember that we are interested in *causal* explanations that exhibit a set of *contributory causes* that together form a *sufficient cause* that entails the violation of the safety property. Hence the formalisation of explanations depends on what we consider as a contributory cause. Different definitions of contributory causes put the spotlight on different aspects of the behaviour, and may blame the safety violation on different portions of a run, as illustrated by subsequence explanations and choice explanations introduced in Sections 4.3 and 5, respectively.

4.2. Contrastive explanations

It has been argued that human beings are often seeking *contrastive* explanations of the form “why has X happened although I expected Y ?” [4], for instance, “why has the accident happened although the car should have kept a safety distance of 2?”. Our goal is thus to explain the violation of a safety property P possibly in contrast to a desired safety property Q . Similar to the non-contrastive case, we do this by constructing an ordered set of contributory causes that bring the system from the set of states that invariantly satisfy Q to the set of states in which the violation of P is inevitable.

We formalise contrastive explanations as follows. As before, satisfaction or violation of Q are tracked by an observer automaton as for P .

Definition 10 (Contrastive explanation). Consider a model $\mathcal{A}$ and two observers $\mathcal{O}_P$ and $\mathcal{O}_Q$ such that $\text{safeLang}_{\mathcal{A}, \emptyset}(\mathcal{O}_Q) \subseteq \text{safeLang}_{\mathcal{A}, \emptyset}(\mathcal{O}_P)$. A *contrastive explanation* with respect to $\langle \mathcal{O}_Q, \mathcal{O}_P \rangle$ for a trace tr is a pair $\langle \pi, \varepsilon \rangle$ where $\pi = \pi'.e$ with $\pi' \in \text{safeLang}_{\mathcal{A}, \emptyset}(\mathcal{O}_Q)$, $\pi \notin \text{safeLang}_{\mathcal{A}, \emptyset}(\mathcal{O}_Q)$, and $\exists tr_1 \exists \nu_0 \exists tr_2 : tr = tr_1.tr_2 \wedge \rho_{\mathcal{A}}(tr_1) = \rho_{\mathcal{A}}(\pi) = \nu_0 \wedge \varepsilon \in \mathcal{E}(\mathcal{A}, \mathcal{O}_P, \nu_0, tr_2)$.

The entities of Definition 10 are sketched in Fig. 4. Intuitively, a contrastive explanation with respect to $\langle \mathcal{O}_Q, \mathcal{O}_P \rangle$ consists of a prefix π after which the violation of Q is inevitable, and an explanation ε for the subsequent violation of P .

The semantics of a contrastive explanation $\langle \pi, \varepsilon \rangle$ is straight-forwardly defined as $\text{sem}_{\mathcal{A}}(\langle \pi, \varepsilon \rangle) = \{\pi.w \mid w \in \text{sem}_{\mathcal{A}}(\varepsilon)\}$.

Example 2 (Running example). If we now consider to contrast the violation of the safety property $P = (sd > 0)$ to a contrastive safety property $Q = (sd \geq 3)$, i.e. a desirable goal of a safety distance of length 3, we must build the observer $\mathcal{O}_Q = \langle \Sigma_{\mathcal{O}_Q}, X_{\mathcal{O}_Q} \cup \{q\}, \text{inv}_{\mathcal{O}_Q}, \delta_{\mathcal{O}_Q}, \nu_{\mathcal{O}_Q}^0, \text{ko}_{\mathcal{O}_Q} \rangle$ following Definition 8 in a similar way as observer $\mathcal{O}_P$ in Section 3.4:

- $\Sigma_{\mathcal{O}_Q} = \{\text{meas}\}$
- $X_{\mathcal{O}_Q} = \{sd, fcs, rcs, qko\}$
- $\text{inv}_{\mathcal{O}_Q} = \text{true}$
- Transitions:
 - $\delta_{\mathcal{O}_Q}(\text{meas}) = (\text{true}, \{ qko \leftarrow \text{true}, \text{ if } (\neg qko \wedge sd + rcs - fcs < 2) \})$
- $\nu_{\mathcal{O}_Q}^0 = (qko = \text{false}, fcs = 1, rcs = 0, sd = 3)$
- $\text{ko}_{\mathcal{O}_Q} = qko$

The contrastive two-car problem becomes: what kind of explanations can we provide about the violation of the safety property $P = (\text{always } sd > 0)$ in contrast to $Q = (sd \geq 3)$ based on the log $\mathcal{L}$, the model $\mathcal{A}$ and the observers $\mathcal{O}_P$ and $\mathcal{O}_Q$?

4.3. Subsequence explanations

As a first instantiation of the explanation function $\mathcal{E}$ introduced above, we investigate in this section a very intuitive type of explanation of a safety violation based on subsequences of traces as proposed in [19]. More specifically, the explanation is aimed at retaining, as contributory causes, only the events relevant to the violation of a safety property P .

A subsequence of a sequence of events $v \in \Sigma^*$ is a sequence of events $u \in \Sigma^*$ such that there exists a monotone function $\psi : [0, |u| - 1] \rightarrow [0, |v| - 1]$ such that $u = [v(\psi(i))]_{i \in [0, |u| - 1]}$. In words, u is obtained from v by erasing part of the events. We write $u \sqsubseteq v$ when u is a subsequence of v .

In the sequel we assume that the log $\mathcal{L}$ is consistent with some trace that violates P .

Definition 11 (Subsequence explanation). Given a behavioural model $\mathcal{A}$, an observer $\mathcal{O}_P$, an initial state v_0 , and a trace $tr \in L^F(\mathcal{A} \parallel \mathcal{O}_P, v_0)$, the set of subsequence explanations for the violation of P is the language provided by the following *SSE* explanation function:

$$SSE(\mathcal{A}, \mathcal{O}_P, v_0, tr) = \min_{\sqsubseteq} \{w \in L^F(\mathcal{A} \parallel \mathcal{O}_P, v_0) \mid w \sqsubseteq tr\}$$

Example 3 (Running example). Consider the two-car system and the run r_2 of $[\mathcal{A} \parallel \mathcal{O}_P]$ starting in v^0 and composed of 25 transitions, t_1 to t_{25} , detailed in Table 3. In this table, the states v_i are denoted with their respective valuation in the following order (*cm, fcs, rcs, sd, sfm, smd, pko*). Columns 3 and 6 should be ignored for now.

The run r_2 starts from the initial state v^0 , where the front car is at speed 1, the rear car does not move and the initial distance is 3. In this run, the front car first stops ($t_1 : v_0 \xrightarrow{fbr} v_1$) while the rear car starts accelerating ($t_2 : v_1 \xrightarrow{ra} v_2$) so the distance between both cars becomes 2 ($t_3 : v_2 \xrightarrow{meas} v_3$). After accelerating and breaking again, the systems returns to the initial state v^0 . The front car then accelerates to keep maximal speed ($t_7 : v_0 \xrightarrow{fa} v_6$) then cruises, breaks and stops ($t_{17} : v_{15} \xrightarrow{fbr} v_{16}$). The rear car, while at distance 2, stops ($t_5 : v_4 \xrightarrow{rbr} v_5$) and then let the front car increase the distance before accelerating twice ($t_{11} : v_9 \xrightarrow{ra} v_{10}$ and $t_{15} : v_{13} \xrightarrow{ra} v_{14}$) to reach the maximal speed in order to reduce the distance. As the sensor has failed ($t_{12} : v_{10} \xrightarrow{fail} v_{11}$), the measured distance is incorrect and is maximal ($t_{13}, \dots, t_{18}$) or even saturated ($t_{14}, \dots, t_{16}$). As the distance keeps reducing, the rear car starts breaking but it is too late; the rear car bumps the front car and the safety property $sd > 0$ is violated ($t_{25} : v_{23} \xrightarrow{meas} v_{24}$).

The run r_2 is associated with the following trace (cf. column 4 of Table 3):

$$tr(r_2) = fbr.ra.meas.fa \dots rbr.meas \in L^F(\mathcal{A} \parallel \mathcal{O}_P).$$

Looking at Table 3, it can be noticed that firing t_6 leads back to the initial state and therefore the sequence from t_7 to t_{25} is also a run of the system that we denote r_3 . As a consequence, $tr(r_3) \sqsubseteq tr(r_2)$. By construction of the model, there is no other trace smaller than $tr(r_3)$ that is in $L^F(\mathcal{A} \parallel \mathcal{O}_P)$ and that is also a subsequence of $tr(r_2)$, it follows that there is only one subsequence explanation associated with trace $tr(r_2)$. Hence we have

$$SSE(\mathcal{A}, \mathcal{O}_P, v^0, tr(r_2)) = \{tr(r_3)\} = \{fa.rcr \dots rbr.meas\},$$

where $tr(r_3)$ traces a sequence of events leading to the violation of P . Note however that the real problem starts at transition $t_{11} : v_9 \xrightarrow{ra} v_{10}$ with the acceleration of the rear car followed by the sensor failure at $t_{12} : v_{10} \xrightarrow{fail} v_{11}$, which then misleads the rear car. So the events related to the initial transitions ($t_7, \dots, t_{10}$) of the subsequence explanation $tr(r_3)$ do not bring any information with respect to the violation of the safety property.

Now let us try to define semantics for a subsequence explanation $\varepsilon \in SSE(\mathcal{A}, \mathcal{O}_P, v^0, tr)$ so that it is complete and semantics-precise. Completeness requires

$$\{tr\} \subseteq sem_{\mathcal{A}}(\varepsilon).$$

The following theorem discusses possible options for the definition of $sem_{\mathcal{A}}$ on a concrete example.

Theorem 1. *There is no semantics for SSE that is complete and semantics-precise.*

Proof. We define the model $\mathcal{A}$ with initial state v_0 , $L(\mathcal{A}) = ab^* \cup ba^* \cup c^+$, and observers $\mathcal{O}$, $\mathcal{O}_{P_1}$, and $\mathcal{O}_{P_2}$ with $L^F(\mathcal{A} \parallel \mathcal{O}) = ab^* \cup ba^*$, $L^F(\mathcal{A} \parallel \mathcal{O}_{P_1}) = ab^*$, and $\mathcal{O}_{P_2}$ with $L^F(\mathcal{A} \parallel \mathcal{O}_{P_2}) = ba^*$. Therefore $L^F(\mathcal{A} \parallel \mathcal{O}_{P_1} \parallel \mathcal{O}_{P_2}, v_0) = \emptyset$.

By Definition 11, $\forall w \in ab^* \cup ba^+$, $\varepsilon a \varepsilon \in SSE(\mathcal{A}, \mathcal{O}, w)$, and $\forall w \in ba^* \cup ab^+$, $\varepsilon b \varepsilon \in SSE(\mathcal{A}, \mathcal{O}, w)$. Completeness with respect to $\mathcal{O}$ implies that

$$ab^* \cup ba^+ \subseteq sem_{\mathcal{A}}(\varepsilon a \varepsilon) \wedge ba^* \cup ab^+ \subseteq sem_{\mathcal{A}}(\varepsilon b \varepsilon)$$

For all semantics satisfying this constraint, $ba \in sem_{\mathcal{A}}(\varepsilon a \varepsilon) \cap sem_{\mathcal{A}}(\varepsilon b \varepsilon) \neq \emptyset$. It follows that $(SSE, sem_{\mathcal{A}})$ is not semantics-precise.

□

The difficulty in defining a complete and precise semantics reflects a more fundamental issue with subsequence explanations: they are merely “shortcuts” to the safety violation, whose corresponding runs are generally incomparable with the actual run.

Besides this issue, another shortcoming of subsequence explanations is the causal relevance of the events in the explanation. If we have a system that systematically performs an initialisation then all the runs — and therefore also the subsequence explanations

Table 3

Run $r_2 = t_1 \dots t_{25} \in \rho([A \parallel \mathcal{O}_P])$ that violates safety property $P : sd > 0$. Note that the six first transitions cycle from and to state v^0 .

	v_i (<i>cm, fcs, rcs, sd, sfm, smd, pko</i>)	$\xi(v_i)$	e_{i+1}	v_{i+1}	$\xi(v_{i+1})$
t_1	$v^0 : (0, 1, 0, 3, 1, 3, 0)$	$\langle 1, 6 \rangle$	<i>fbr</i>	$v_1 : (1, 0, 0, 3, 1, 3, 0)$	$\langle 1, 7 \rangle$
t_2	$v_1 : (1, 0, 0, 3, 1, 3, 0)$	$\langle 1, 7 \rangle$	<i>ra</i>	$v_2 : (2, 0, 1, 3, 1, 3, 0)$	$\langle 1, 7 \rangle$
t_3	$v_2 : (2, 0, 1, 3, 1, 3, 0)$	$\langle 1, 7 \rangle$	<i>meas</i>	$v_3 : (0, 0, 1, 2, 1, 2, 0)$	$\langle 1, 7 \rangle$
t_4	$v_3 : (0, 0, 1, 2, 1, 2, 0)$	$\langle 1, 7 \rangle$	<i>fa</i>	$v_4 : (1, 1, 1, 2, 1, 2, 0)$	$\langle 1, 6 \rangle$
t_5	$v_4 : (1, 1, 1, 2, 1, 2, 0)$	$\langle 1, 6 \rangle$	<i>rbr</i>	$v_5 : (2, 1, 0, 2, 1, 2, 0)$	$\langle 1, 6 \rangle$
t_6	$v_5 : (2, 1, 0, 2, 1, 2, 0)$	$\langle 1, 6 \rangle$	<i>meas</i>	$v^0 : (0, 1, 0, 3, 1, 3, 0)$	$\langle 1, 6 \rangle$
t_7	$v^0 : (0, 1, 0, 3, 1, 3, 0)$	$\langle 1, 6 \rangle$	<i>fa</i>	$v_6 : (1, 2, 0, 3, 1, 3, 0)$	$\langle 1, 6 \rangle$
t_8	$v_6 : (1, 2, 0, 3, 1, 3, 0)$	$\langle 1, 6 \rangle$	<i>rcr</i>	$v_7 : (2, 2, 0, 3, 1, 3, 0)$	$\langle 1, 6 \rangle$
t_9	$v_7 : (2, 2, 0, 3, 1, 3, 0)$	$\langle 1, 6 \rangle$	<i>meas</i>	$v_8 : (0, 2, 0, 5, 1, 5, 0)$	$\langle 1, 7 \rangle$
t_{10}	$v_8 : (0, 2, 0, 5, 1, 5, 0)$	$\langle 1, 7 \rangle$	<i>fcr</i>	$v_9 : (1, 2, 0, 5, 1, 5, 0)$	$\langle 1, 7 \rangle$
t_{11}	$v_9 : (1, 2, 0, 5, 1, 5, 0)$	$\langle 1, 7 \rangle$	<i>ra</i>	$v_{10} : (2, 2, 1, 5, 1, 5, 0)$	$\langle 1, 6 \rangle$
t_{12}	$v_{10} : (2, 2, 1, 5, 1, 5, 0)$	$\langle 1, 6 \rangle$	<i>fail</i>	$v_{11} : (2, 2, 1, 5, 0, 5, 0)$	$\langle 1, 5 \rangle$
t_{13}	$v_{11} : (2, 2, 1, 5, 0, 5, 0)$	$\langle 1, 5 \rangle$	<i>meas</i>	$v_{12} : (0, 2, 1, 6, 0, 10, 0)$	$\langle 1, 5 \rangle$
t_{14}	$v_{12} : (0, 2, 1, 6, 0, 10, 0)$	$\langle 1, 5 \rangle$	<i>fbr</i>	$v_{13} : (1, 1, 1, 6, 0, 10, 0)$	$\langle 1, 4 \rangle$
t_{15}	$v_{13} : (1, 1, 1, 6, 0, 10, 0)$	$\langle 1, 4 \rangle$	<i>ra</i>	$v_{14} : (2, 1, 2, 6, 0, 10, 0)$	$\langle 1, 4 \rangle$
t_{16}	$v_{14} : (2, 1, 2, 6, 0, 10, 0)$	$\langle 1, 4 \rangle$	<i>meas</i>	$v_{15} : (0, 1, 2, 5, 0, 10, 0)$	$\langle 1, 4 \rangle$
t_{17}	$v_{15} : (0, 1, 2, 5, 0, 10, 0)$	$\langle 1, 4 \rangle$	<i>fbr</i>	$v_{16} : (1, 0, 2, 5, 0, 10, 0)$	$\langle 1, 3 \rangle$
t_{18}	$v_{16} : (1, 0, 2, 5, 0, 10, 0)$	$\langle 1, 3 \rangle$	<i>rcr</i>	$v_{17} : (2, 0, 2, 5, 0, 10, 0)$	$\langle 1, 3 \rangle$
t_{19}	$v_{17} : (2, 0, 2, 5, 0, 10, 0)$	$\langle 1, 3 \rangle$	<i>meas</i>	$v_{18} : (0, 0, 2, 3, 0, 6, 0)$	$\langle 1, 3 \rangle$
t_{20}	$v_{18} : (0, 0, 2, 3, 0, 6, 0)$	$\langle 1, 3 \rangle$	<i>fcr</i>	$v_{19} : (1, 0, 2, 3, 0, 6, 0)$	$\langle 1, 2 \rangle$
t_{21}	$v_{19} : (1, 0, 2, 3, 0, 6, 0)$	$\langle 1, 2 \rangle$	<i>rcr</i>	$v_{20} : (2, 0, 2, 3, 0, 6, 0)$	$\langle 0, 1 \rangle$
t_{22}	$v_{20} : (2, 0, 2, 3, 0, 6, 0)$	$\langle 0, 1 \rangle$	<i>meas</i>	$v_{21} : (0, 0, 2, 1, 0, 2, 0)$	$\langle 0, 1 \rangle$
t_{23}	$v_{21} : (0, 0, 2, 1, 0, 2, 0)$	$\langle 0, 1 \rangle$	<i>fcr</i>	$v_{22} : (1, 0, 2, 1, 0, 2, 0)$	$\langle 0, 0 \rangle$
t_{24}	$v_{22} : (1, 0, 2, 1, 0, 2, 0)$	$\langle 0, 0 \rangle$	<i>rbr</i>	$v_{23} : (2, 0, 1, 1, 0, 2, 0)$	$\langle 0, 0 \rangle$
t_{25}	$v_{23} : (2, 0, 1, 1, 0, 2, 0)$	$\langle 0, 0 \rangle$	<i>meas</i>	$v_{24} : (0, 0, 1, 0, 0, 0, 1)$	$\langle 0, 0 \rangle$

— share a common prefix, even though it is debatable whether this initialisation should be considered as a contributory cause. The problem of non informative events has been pointed out for the two-car running example in [Example 3](#).

Moreover, subsequence explanations only consider events and leave aside the configuration of variables which can however provide important explanatory information.

However, note that subsequence explanations are well suited to compute a concise contrastive prefix π as part of a contrastive explanation.

Choice explanations, that we propose in the next section, avoid the downsides of subsequence explanations by adopting a different characterisation of contributory causes.

5. Choice explanations

In this section, we introduce a type of explanations where the contributory causes are *choices* of the system. For this purpose, we define the notion of *level of choice*, which can be seen as a generalisation of the concept of *fate and free will* of [\[3\]](#). Throughout this section, the inputs to the explanation problem are a model $\mathcal{A}$, an observer $\mathcal{O}_P$ for the violation of a safety property P , and a run $r \in \rho(\mathcal{A} \parallel \mathcal{O}_P, v_0)$.

5.1. The level of choice

The notion of level of choice is formally defined over any semantic graph whose alphabet Σ is partitioned into a set of controllable events Σ_c and a set of uncontrollable events Σ_u .

Definition 12 (Level of choice). Given the semantic graph $G = \langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle := [\mathcal{A} \parallel \mathcal{O}_P]$ such that $\Sigma = \Sigma_c \uplus \Sigma_u$, we say that the function $\xi : \mathcal{V} \rightarrow (\mathbb{N} \cup \{\infty\})^2$ is the level of choice of G if:

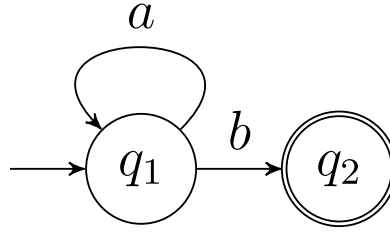
1. $\forall v \in \mathcal{V}^F : \xi(v) = \langle 0, 0 \rangle$
2. $\forall v \in \mathcal{V}$ such that $\mathcal{V}^F$ is not reachable from v , $\xi(v) = \langle \infty, \infty \rangle$
3. for all other states $v \in \mathcal{V}$

$$\xi(v) = \langle \min\{\xi_c(v), \xi_u(v)\}, \xi_{cu}(v) \rangle$$

with:

$$\xi_c(v) := \min^+ \{ \xi_1(v') \mid \exists e \in \Sigma_c : v \xrightarrow{e} v' \}$$

$$\xi_u(v) := \min \{ \xi_1(v') \mid \exists e \in \Sigma_u : v \xrightarrow{e} v' \}$$

Fig. 5. LTS motivating the maximality requirements on ξ .

$$\xi_{cu}(v) := \min^+(\{\xi_2(v') \mid \exists e \in \Sigma : v \xrightarrow{e} v'\})$$

where for any set S :

$$\min^+(S) = \begin{cases} \min(S) & \text{if } |S| \leq 1 \\ 1 + \min(S) & \text{otherwise} \end{cases}$$

and we set $\min(\emptyset) = \infty$.

4. **Maximality:** ξ is in both components maximal among the functions fulfilling the preceding conditions, i.e. $\forall \xi' : \mathcal{V} \rightarrow (\mathbb{N} \cup \{\infty\})^2$ fulfilling the preceding conditions, $\forall v \in \mathcal{V}, \xi_1'(v) \leq \xi_1(v) \wedge \xi_2'(v) \leq \xi_2(v)$.

The level of choice $\xi(v) = \langle \ell_1, \ell_2 \rangle$ has two components. ℓ_1 is qualified as the *controllable level of choice* as it indicates how many “bad choices” among controllable events still separate a state s from the states in $\mathcal{V}_F$ in the worst case, that is, for a hostile environment firing uncontrollable transitions. In contrast, ℓ_2 is qualified as the *closed-system level of choice* as it is the number of remaining “bad choices” without distinguishing controllable and uncontrollable events. ℓ_2 amounts to the level of choice of [19]. Both are instrumental in constructing choice explanations.

Intuitively, considering the model $\mathcal{A}$ and the safety property P , we first compute the level of the choice of the semantic graph $[\mathcal{A} \parallel \mathcal{O}_P] = \langle \mathcal{V}, \Sigma, \rightarrow, \nu^0, \mathcal{V}^F \rangle$ of $\mathcal{A} \parallel \mathcal{O}_P$. The runs of the behavioural model $\mathcal{A}$ violating the safety property are then represented by runs of $\mathcal{A} \parallel \mathcal{O}_P$ that end in an accepting state, i.e. a state in $\mathcal{V}^F$. We will use level ℓ_2 to identify uncontrollable events that challenge the system by moving it closer to the violation of P , and level ℓ_1 to identify system events that do the same in spite of the existence of a “safer” alternative.

The first condition of Definition 12 requires that the level of choice of accepting states is 0, which means that no choices can be made in those states to violate the safety property because the violation already happened. The second condition requires the level of choice of the states from which $\mathcal{V}^F$ is not reachable to be ∞ ; this means that in those states, no choice can be made to reach a violation. The third condition is different for the controllable level of choice ℓ_1 and the closed-system level of choice ℓ_2 . Given a state v , let us define the “controllable postset” v_c^* and the “uncontrollable postset” v_u^* as the set of successors via controllable transitions and uncontrollable transitions respectively. Then:

- the closed-system level of choice $\xi_2(v)$ is determined considering indifferently successors in the controllable postset v_c^* and the uncontrollable postset v_u^* . If the level of choice of the successors is uniform, then the level of choice of v is the same as its successors because the decision about the out-going transition does not impact the level of choice. On the other hand, if the level of choice of the successors is not uniform, then the level of choice of v is equal to the increment by one of the lowest level of choice in $v_c^* \cup v_u^*$. It indeed means that an impacting event has been issued by the system or the environment at v .
- The controllable level of choice $\xi_1(v)$ only accounts for decisions of the system, independently of those of the environment. Indeed, we have $\xi_1 \leq \xi_u$ where $\xi_u(v)$ is the minimum level of choice ξ_1 in v_u^* . Therefore, the occurrence of an uncontrollable event does not decrease ξ_1 . ξ_c is the counterpart to ξ_{cu} , it depends on v_c^* and ξ_1 instead of v^* and ξ_2 . The controllable level of choice $\xi_1(v)$ is finally obtained as the minimum of both values.

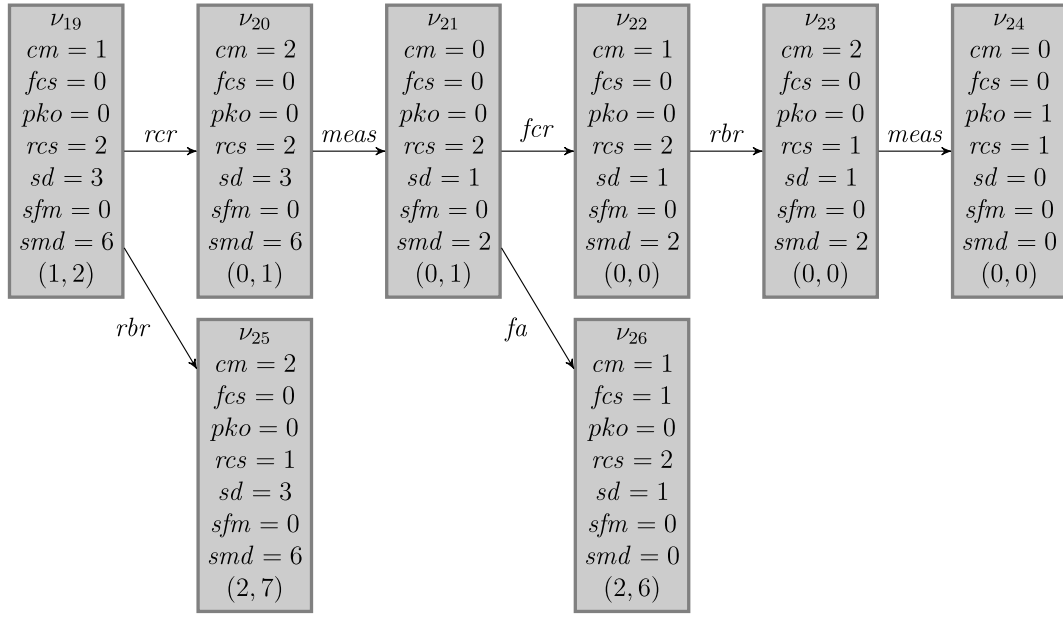
The fourth requirement (maximality) ensures that the levels of choice are not unnecessarily pessimistic. The LTS of Fig. 5, where both events a and b are controllable, illustrates this point. The requirements of Definition 12, excluding the maximality requirement, allow for two levels of choice: $q_2 \mapsto \langle 0, 0 \rangle$ and either $q_1 \mapsto \langle 0, 0 \rangle$ or $q_1 \mapsto \langle 1, 1 \rangle$. Only the second one correctly represents the fact that in q_1 , the system can still avoid entering q_2 .

Lemma 1 (Unicity of the level of choice). *The level of choice as defined in Definition 12 is unique.*

Proof. Each of the functions ξ_1 and ξ_2 is computed as a greatest fixpoint, where for each level $j \geq 1$, the set of states v with $\xi_i(v) = j$ is uniquely determined by the set of states v' with $\xi_i(v') \leq j - 1$, $i = 1, 2$. It follows that the greatest fixpoint is unique. $\square$

Example 4 (Running example). Fig. 6 represents part of $[\mathcal{A} \parallel \mathcal{O}_P]$ where each state is associated with its level of choice.

By construction of the model, from every state v there exists at least one run that reaches a state that violates P : $sd > 0$, so there is no state v such that $\xi(v) = \langle \infty, \infty \rangle$. The set $\mathcal{V}^F$ is composed of 3 states in all of which the sensor is faulty ($sfm = 0$) and the rear car ran out of time to reduce its speed given the speed of the front car fcs and the actual distance between the two cars sd .

Fig. 6. Part of $[A||O_P]$ with levels of choice.

The level of choice of the top left state v_{19} is $\xi(v_{19}) = (1, 2)$. Its controllable level of choice $\xi_1(v_{19}) = 1$ indicates that from this state, there is precisely one occurrence of a controllable event that leads to a state where the rear car no longer has any available controllable events to maintain the safety property $sd > 0$. If the rear car opts for this controllable event, which is rcr , the safety property is compromised, depending entirely on the actions of the front car. The closed-system level of choice $\xi_2(v_{19}) = 2$ means that the rear car will eventually violate the safety property as soon as two specific controllable/uncontrollable events occur. Fig. 6 illustrates that the controllable event rcr is the bad choice as it leads to the state v_{20} where the level of choice drops to $\xi(v_{20}) = (0, 1)$. After the occurrence of rcr , no further controllable events are possible. The second event that leads to the violation of the safety property is fcr , which is uncontrollable. After its occurrence, the system reaches the state v_{22} with level of choice $\xi(v_{22}) = (0, 0)$ from which the violation of $sd > 0$ is inevitable.

Algorithm 1 the level of choice $\xi = \langle \xi_1, \xi_2 \rangle$. Both components of ξ can be computed independently using a variant of Dijkstra's algorithm [21] for single-source shortest paths.

Algorithm 1: Computation of $\xi_i, i \in \{1, 2\}$ on $G = (\mathcal{V}, \rightarrow)$.

```

1  initialise  $(\chi_i)$ 
2   $S := \mathcal{V}$ 
3  while  $S \neq \emptyset$  do
4     $\mu := \text{extractMin}_{\chi_i}(S)$ 
5    for predecessor  $v \xrightarrow{e} \mu$  do
6       $\chi_i(v) := \min\{\chi_i(v), \text{update}\chi_i(v \xrightarrow{e} \mu)\}$ 
7   $\xi_i := \chi_i$ 
8  return  $\xi_i$ 

9  function  $\text{update}\chi_1(v \xrightarrow{e} \mu)$ 
10   if  $e \in \Sigma^c$  then
11      $h := \min^+(\{\chi_1(v') \mid \exists e' \in \Sigma^c : v \xrightarrow{e'} v'\})$ 
12   else
13      $h := \chi_1(\mu)$ 
14   return  $h$ 

15 function  $\text{update}\chi_2(v \xrightarrow{e} \mu)$ 
16   return  $\min^+(\{\chi_2(v') \mid \exists e' \in \Sigma : v \xrightarrow{e'} v'\})$ 

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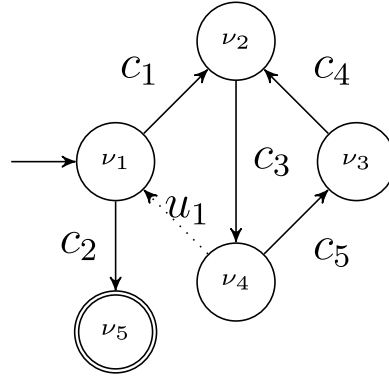


Fig. 7. Small example illustrating Algorithm 1.

Algorithm 1 is illustrated based on the simple semantic graph of Fig. 7. In this example only ν_5 is an accepting state. Events c_i 's are controllable events and only u_1 is uncontrollable.

Algorithm 1 computes ξ_1 as follows. After the initialisation (Line 1), the current level of choice of any accepting state v is $\chi_1(v) = 0$ and for any other state v' , it is set to $\chi_1(v') = +\infty$. In this example, $\chi_1(\nu_5) = 0$ and $\chi_1(\nu_i) = +\infty$, $1 \leq i \leq 4$. The algorithm then consists in storing any state of G in a set S and in iterating over the set S . At each iteration, $\text{extractMin}_{\chi_1}(S)$ (Line 4) extracts from S a state μ whose level of choice $\chi_1(\mu)$ is minimal among the levels of choice of all states in S . The first extraction from S is thus $\mu = \nu_5$ (Line 4) for which Line 7 already provides $\xi_1(\nu_5) = \chi_1(\nu_5) = 0$. State ν_5 has one predecessor through a controllable transition, that is ν_1 , which is considered next (Line 6). State ν_1 has two controllable output transitions leading to ν_2 and ν_5 , it follows from Line 11 that $h = \min^+(\chi_1(\nu_5), \chi_1(\nu_2)) = \min^+(0, +\infty) = 1$, and from Line 6 that $\chi_1(\nu_1) = \min(\chi_1(\nu_1), h) = \min(+\infty, 1) = 1$.

The set S now contains $\{\nu_1, \nu_2, \nu_3, \nu_4\}$ and ν_1 is the only state whose level of choice $\chi_1(\nu_1)$ is minimal among the levels of choice of all states in S . The second extraction is therefore $\mu = \nu_1$. At this point, one already knows that the final level of choice of ν_2 cannot be 0 because the current minimal one in S is greater than 0, given by $\chi_1(\nu_1) = 1 = \xi_1(\nu_1)$. The only predecessor of ν_1 is ν_4 through an uncontrollable transition labeled u_1 . Line 13 then enforces $h = \chi_1(\nu_1) = 1$, and Line 6 provides $\chi_1(\nu_4) = \min(\chi_1(\nu_4), h) = \min(+\infty, 1) = 1$.

The set S now contains $\{\nu_2, \nu_3, \nu_4\}$ and ν_4 is the only state whose level of choice $\chi_1(\nu_4)$ is minimal among the levels of choice of all states in S . At this point, one already knows that the level of choice of ν_3 cannot be lower than 1, which is the min of the states in S . Next extraction is hence $\mu = \nu_4$. The only predecessor of ν_4 is ν_2 through a controllable transition. State ν_2 has only one controllable output transition leading to ν_4 , it follows from Line 11 that $h = \min^+(\chi_1(\nu_4)) = 1$, and from Line 6 that $\chi_1(\nu_2) = \min(\chi_1(\nu_2), h) = \min(+\infty, 1) = 1$.

The set S now contains $\{\nu_2, \nu_3\}$ and ν_2 is the only state whose level of choice $\chi_1(\nu_2)$ is minimal among the levels of choice of all states in S . ν_2 is hence extracted. State ν_2 has two predecessors ν_3 and ν_1 , both through controllable transitions. ν_3 has only one output controllable transition to ν_2 , hence for the same reasons as before $h = \min^+(\chi_1(\nu_2)) = 1$ and $\chi_1(\nu_3) = 1$. On the other hand, as already seen, ν_1 has two output controllable transitions to ν_2 and ν_5 . From Line 11, $h = \min^+(\chi_1(\nu_5), \chi_1(\nu_2)) = \min^+(0, 1) = 1$, which leaves $\chi_1(\nu_1)$ unchanged and equal to 1.

$\mu = \nu_3$ is extracted next from S . It has only ν_4 as predecessor. It follows from Line 11 that $h = \min^+(\chi_1(\nu_3)) = 1$ and from Line 6 that $\chi_1(\nu_4) = \min(\chi_1(\nu_4), h) = \min(1, 1) = 1$, which also leaves the value unchanged.

Finally, Line 7 provides $\xi_1(\nu_i) = \chi_1(\nu_i)$, $i = 1, \dots, 5$, hence $\xi_1(\nu_1) = 1$, $\xi_1(\nu_2) = 1$, $\xi_1(\nu_3) = 1$, $\xi_1(\nu_4) = 1$, and $\xi_1(\nu_5) = 0$, which is returned by Algorithm 1 (Line 8).

When Algorithm 1 computes ξ_2 on this example, the extraction order is similar and it starts by computing $\chi_2(\nu_5) = 0$, $\chi_2(\nu_1) = 1$. The main difference is when ν_1 is extracted. The predecessor of ν_1 is ν_4 , that has two output transitions to ν_1 and ν_3 . Independently of the status of the transition between ν_1 and ν_4 , Line 16 enforces $\chi_2(\nu_4) = \min^+(\chi_2(\nu_1), \chi_2(\nu_3)) = \min^+(1, +\infty) = 2$. The final result is $\xi_2(\nu_5) = 0$, $\xi_2(\nu_1) = 1$, and $\xi_2(\nu_2) = \xi_2(\nu_3) = \xi_2(\nu_4) = 2$.

5.2. Algorithm correctness

As illustrated in the previous example, the correctness of Algorithm 1 relies on the property that any state μ that is extracted from S is always such that $\chi_i(\mu) = \xi_i(\mu)$ when it is extracted and $\chi_i(\mu)$ remains unchanged till the end of the algorithm. This section is devoted to the proof of this result. In the following, $\mathbf{P}(n)$, $n \geq 0$ denotes the following property: let $\ell_{\min}(n)$ be the minimal level of choice associated with a state of S at Line 4 before the $(n+1)^{\text{th}}$ extraction,

$$\forall \mu \in S, \chi_i(\mu) = \ell_{\min}(n) \Rightarrow \chi_i(\mu) = \xi_i(\mu), \quad (8)$$

Lemma 2. Property $\mathbf{P}(0)$ holds.

Proof. After the initialisation of χ_i (Line 1), the level of choice of any state $v \in \mathcal{V}$ is either set to 0 (v violates P) or $+\infty$ (v does not violate P) so $\chi_i(v) \geq 0$. If $\ell_{\min}(n) = 0$ before the first extraction, then any state μ of S with $\chi_i(\mu) = 0$ violates P so $\chi_i(\mu) = \xi_i(\mu) = 0$ (see Definition 12 item 1). If $\ell_{\min}(n) = +\infty$, there is no state that violates P . Any state of $S = \mathcal{V}$ is such that $\xi_i(\mu) = \chi_i(\mu) = +\infty$ (see Definition 12 item 2). Hence $\mathbf{P}(0)$. $\square$

Lemma 3. For all $n \in [0, |\mathcal{V}| - 1]$, $\mathbf{P}(n) \Rightarrow \mathbf{P}(n + 1)$.

Proof. Let $n \in [0, |\mathcal{V}| - 1]$ and suppose $\mathbf{P}(n)$ is true. Let μ be the $n + 1$ th extraction from S . If $\chi_i(\mu) = +\infty$, any state s of S is such that $\chi_i(s) = +\infty$ and none of the level of choice will be updated. Any state s of S is such that $\xi_i(s) = \chi_i(s) = +\infty$, hence the result $\mathbf{P}(n + 1)$. Suppose now that $\chi_i(\mu) = \ell < +\infty$, and consider any transition $v \xrightarrow{e'} \mu$. Four distinct conditions over $\chi_i(v)$ must be analyzed.

1. $\chi_i(v) = +\infty$:
 - Suppose $v \xrightarrow{e'} \mu$ is uncontrollable. If $i = 1$, $\chi_i(v) := \ell$ and if $i = 2$, $\chi_i(v) := \ell$ or $\chi_i(v) := \ell + 1$;
 - Suppose $v \xrightarrow{e'} \mu$ is controllable, it follows in any case $\chi_i(v) := \ell$ or $\chi_i(v) := \ell + 1$.
2. $\chi_i(v) \leq \ell$: the value of $\chi_i(v)$ remains unchanged due to Line 6.
3. $\chi_i(v) = \ell + 1$: same as case $\chi_i(v) = +\infty$.
4. $\chi_i(v) > \ell + 1 < +\infty$: this case is impossible. For $\chi_i(v) = \ell' > \ell + 1$, $\ell' < +\infty$, it would mean that $\chi_i(v)$ has been updated at least once before the current extraction μ and this previous extraction μ' would be a successor of v with $\chi_i(\mu') \in \{\ell' - 1, \ell'\} > \ell$.

Finally, for any predecessor v of μ , $\chi_i(v) \leq \ell + 1$ whether $\chi_i(v)$ was updated or not after the $n + 1$ th extraction. Moreover any predecessor v such that $\chi_i(v) = \ell + 1$ must be still in S . After the update of all the predecessors of μ , let us analyze the possible cases based on the following distinct conditions over the new value of $\ell_{\min}(n)$:

1. $\ell_{\min}(n) < \ell$: impossible case. Any effective update is either ℓ or $\ell + 1$. Any state with lower level of choice is not in S .
2. $\ell_{\min}(n) > \ell + 1 < +\infty$: impossible case. A state in S with such a level $\ell_{\min}(n) < +\infty$ should be the predecessor of a state with a level $\ell_{\min}(n)$ or $\ell_{\min}(n) - 1 > \ell$ but no state like that has been extracted yet.
3. $\ell_{\min}(n) = \ell$: There is still at least one state v in S such that $\chi_i(v) = \ell$. Such a state could be there before the current extraction of μ and as $\mathbf{P}(n)$ holds, $\xi_i(v) = \chi_i(v) = \ell$. Otherwise it is a predecessor of μ . As $\xi_i(\mu) = \ell$ and $\chi_i(v) = \ell$ it follows that $\xi_i(v) = \chi_i(v) = \ell$ (see Definition 12 item 3), hence $\mathbf{P}(n + 1)$.
4. $\ell_{\min}(n) = \ell + 1$: it implies that there is no more state μ in S such that $\chi_i(\mu) = \ell$. It follows that there is no more state μ in S such that $\xi_i(\mu) = \ell$. Indeed, if a state μ is such that $\xi_i(\mu) = \ell$, the algorithm would have already assigned $\chi_i(\mu)$ with ℓ , which is contradictory. Now as only states s with $\chi_i(s) \leq \ell$ have been extracted till now, any state $v \in S$ such that $\chi_i(v) = \ell + 1$ must be the predecessor of a state μ' such that $\chi_i(\mu') = \ell$, which means, by $\mathbf{P}(n)$ that $\xi_i(\mu') = \ell$. In the case of $i = 1$, it follows that state v only has output controllable transitions and one at least is leading to a state v' such that $\chi_i(v') > \ell$. If $i = 2$, at least one transition from v is leading to a state v' such that $\chi_i(v') > \ell$. In both case, as $\chi_i(v') > \ell$ now, it implies that $\xi_i(v') \neq \ell$. Therefore $\xi_i(v) = \ell + 1$ (see $\xi_{cu}(v)$ when $i = 1$ and $\xi_{cu}(v)$ when $i = 2$ in Definition 12), hence $\mathbf{P}(n + 1)$.
5. $\ell_{\min}(n) = +\infty$. This case happens when the extracted state μ was the only state in S with $\chi_i(\mu) < +\infty$ and state μ has no predecessors (other than itself). In this case, the other states cannot reach the state that violates P so by definition $\xi_1(s) = +\infty = \chi(s)$ (see Definition 12 item 2). Hence $\mathbf{P}(n + 1)$.

□

Theorem 2. Algorithm 1 is correct in that it computes the intended levels of choice.

Proof. Every state of $G = (\mathcal{V}, \rightarrow)$ is initially stored in S and Algorithm 1 extracts every state μ of S once and only once. Moreover, Lemma 2 and 3 ensure that at extraction time $\xi_i(\mu) = \chi_i(\mu)$. Then notice that as soon as μ is extracted, its level of choice $\chi_i(\mu)$ remains unchanged: any modification would come from a successor of μ with a level of choice greater or equal to l so the update would be greater than l which is impossible due to Line 6. □

5.3. Properties of the level of choice

The level of choice possesses several interesting properties that, as we will explore later, help in developing well-founded explanations.

Lemma 4. Given a model $\mathcal{A}$ and an observer $\mathcal{O}_P$, let ξ be the level of choice function of $\mathcal{A} \parallel \mathcal{O}_P$ and $\langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle := [\mathcal{A} \parallel \mathcal{O}_P]$. We define the set of states from which a violation of the safety property observed by $\mathcal{O}_P$ is inevitable as the smallest set $X \subset \mathcal{V}$ such that $\mathcal{V}^F \subset X$ and $\forall v \in \mathcal{V}, v^* \neq \emptyset \wedge v^* \subset X \implies v \in X$. Then, from any state $v \in \mathcal{V}$ such that $\xi_2(v) = 0$, a violation of the safety property observed by $\mathcal{O}_P$ is inevitable.

Proof. We prove by induction on $i \in \mathbb{N}$ that $\forall v \in \mathcal{V}, \xi_2^i(v) = 0 \implies$ a violation of the safety property observed by $\mathcal{O}_P$ is inevitable from v .

For $i = 0$, let $v \in \mathcal{V}$ such that $\xi_2^0(v) = 0$ it implies by definition of ξ^0 that $v \in \mathcal{V}^F$ and therefore by definition of $\mathcal{V}^F$, v is accepting for the observer.

For $i \in \mathbb{N}$, let $v \in \mathcal{V}$ such that $\xi_2^{i+1}(v) = 0 \wedge \xi_2^i(v) \neq 0$. By definition of ξ^{i+1} we have $\xi_2^{i+1}(v) = \min^+(\{\xi_2^i(v') \mid \exists e \in \Sigma : v \xrightarrow{e} v'\}) = 0 \implies \forall v' \in v^*, \xi_2^i(v') = 0$

Therefore $\forall v' \in v^*, \xi^i(v') = 0 \wedge v^* \neq \emptyset$. We use the induction hypothesis and all runs lead to states where the violations of P is inevitable. Hence a violation of the safety property observed by $\mathcal{O}_P$ is inevitable from v . □

Intuitively, when the system is in a state v in which the level of choice is $\xi(v) = \langle 0, 0 \rangle$, from such a state, the violation is inevitable even if the environment is collaborative.

Proposition 2. Given a model $\mathcal{A}$ and an observer $\mathcal{O}_P$, let ξ be the level of choice of $\langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle := [\mathcal{A} \parallel \mathcal{O}_P]$. We have that:

1. $\forall v \in \mathcal{V} : \xi_1(v) \leq \xi_2(v)$ and, if $\Sigma_u = \emptyset$, $\xi_1(v) = \xi_2(v)$;
2. $\forall v, v' \in \mathcal{V} \forall e \in \Sigma_u : (v \xrightarrow{e} v' \implies \xi_1(v') \geq \xi_1(v))$.
3. $\forall v, v' \in \mathcal{V} \forall e \in \Sigma \forall i \in \{1, 2\} : (v \xrightarrow{e} v' \implies \xi_i(v) - \xi_i(v') \leq 1)$.
4. If $\xi(v) = (\ell_1, \ell_2)$ then any run from v to $\mathcal{V}^F$ encompasses at least ℓ_2 transitions, at least ℓ_1 of which are controllable.

Intuitively, 1. means that the closed-system level of choice is an upper bound for the controllable level of choice, and if all events are controllable, both coincide. Line 2. states that uncontrollable events do not decrease the controllable level of choice, as the definition of ξ_1 anticipates their effect. According to 3., any transition decreasing one of the levels of choice decrements it by at most one. Finally, 4. provides a lower bound on the remaining number of controllable and uncontrollable transitions on any run, starting from a given state, that violates the safety property.

Proof.

1. By induction on the iterations: the claim holds for ξ^0 . Induction step: whenever $\xi_1^i(v) \leq \xi_2^i(v)$ then $\min\{\xi_c^i(v), \xi_u^i(v)\} \leq \xi_{cu}^i(v)$, hence $\xi_1^{i+1}(v) \leq \xi_2^{i+1}(v)$. For $\Sigma_u = \emptyset$, $\xi_1(v) = \xi_2(v)$ follows by equality of $\xi_c^i(v)$ and $\xi_{cu}^i(v)$.
2. We have by Definition 12 that $\xi_1(v) = \min\{\xi_c(v), \xi_u(v)\}$ and $\xi_u(v) = \min\{\xi_1(v'') \mid \exists e \in \Sigma_u : v \xrightarrow{e} v''\} \leq \xi_1(v')$, hence $\xi_1(v) \leq \xi_1(v')$.
3. We have by Definition 12 that $\xi_1(v) = \min\{\xi_c(v), \xi_u(v)\} \leq \min^+(\{\xi_1(v') \mid \exists e \in \Sigma : v \xrightarrow{e} v'\}) \leq 1 + \min(\{\xi_1(v') \mid \exists e \in \Sigma : v \xrightarrow{e} v'\}) \leq 1 + \xi_1(v')$, hence $\xi_1(v') - \xi_1(v) \geq -1$. $\xi_2(v) = \xi_{cu}^i(v) \leq 1 + \min(\{\xi_2(v') \mid \exists e \in \Sigma : v \xrightarrow{e} v'\}) \leq 1 + \xi_2(v')$, hence $\xi_2(v') - \xi_2(v) \geq -1$.
4. The first part of the claim follows directly from item 3. The second part then follows with item 2.

□

5.4. Contributory causes

Given a model $\mathcal{A}$, an observer $\mathcal{O}_P$, and ξ the level of choice of $\langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle := [\mathcal{A} \parallel \mathcal{O}_P]$, consider a transition $v \xrightarrow{e} v'$ of $[\mathcal{A} \parallel \mathcal{O}_P]$.

- Considering the controllable level of choice, if $\xi_1(v') = \xi_1(v) - 1$ the transition drives the system closer to the violation. If $\xi_1(v') = \xi_1(v)$, then the transition does not involve any action from the system that drives it closer to the violation although it may involve actions from the environment that do so. If $\xi_1(v') > \xi_1(v)$ then the transition repairs past actions from the system although actions from the environment may remain unrepaired. Those transitions are relevant to understand the actual actions leading to the violation. If $\xi_1(v') = \infty$, then the run does not violate P and any continuation of the run is also safe with respect to P .
- Considering the closed-system level of choice, if $\xi_2(v') = \xi_2(v) - 1$ the transition involves an action that drives the system closer to the violation. If $\xi_2(v') = \xi_2(v)$ then the transition can be seen as a delay, the violation is only delayed until a transition decreases ξ_2 from a state at the same level $\xi_2(v)$. If $\xi_2(v') > \xi_2(v)$ then the transition repairs past actions. The extreme case is when $\xi_2(v') = \infty$, implying that the rest of the run is safe.

In a run, the impact of an event depends on future events. If there is a loop in the run, then the actions present in the loop are compensated by transitions that increase the level of choice and they might not be related to the violation at the end of the run. The effect of some transitions may not be compensated, however. We now characterise counterfactual causation of a safety violation.

As argued in the introduction, defining causation means focusing one's attention on certain aspects of a model and execution. In the setting of choice explanations, contributory causes take the following form.

Definition 13 (Contributory causes). Given a model $\mathcal{A}$, an observer $\mathcal{O}_P$, a state v_0 of $\mathcal{A} \parallel \mathcal{O}_P$, and a run $r = v_0 \xrightarrow{e_1} v_1 \dots \xrightarrow{e_n} v_n \in \rho(\mathcal{A} \parallel \mathcal{O}_P, v_0)$, let $\langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle := [\mathcal{A} \parallel \mathcal{O}_P]$ and ξ be its level of choice.

A transition $t = v_{i-1} \xrightarrow{e_i} v_i$ of r is a *controllable contributory cause* for the safety violation observed by $\mathcal{O}_P$ if

$$\forall j \in [i, n] : \xi_1(v_{i-1}) > \xi_1(v_j)$$

A transition t is a *closed-system contributory cause* for the safety violation observed by $\mathcal{O}_P$ if

$$\forall j \in [i, n] : \xi_2(v_{i-1}) > \xi_2(v_j)$$

In both cases we call transition t a (controllable and closed-system, respectively) *effective choice transition*. Let $ECT_{\xi_1}(r)$ (resp. $ECT_{\xi_2}(r)$) be the sets of controllable (resp. closed-system) effective choice transitions in r , and let $ECT_{\xi}(r) = ECT_{\xi_1}(r) \cup ECT_{\xi_2}(r)$.

An effective choice transition of a run is a transition that decreases the level of choice, and this decrease is not compensated later in the suffix following the transition. A transition in $ECT_{\xi_1}(r)$ is a bad choice of the system, as it decreases its ability to avoid the violation of the safety property. A transition in $ECT_{\xi_2}(r)$ is a bad choice of the system or a bad action from the environment that drives effectively the system closer to the violation.

Searching for responsibility on the side of a system rather than its environment, we will mostly be interested in explaining violations in terms of controllable contributory causes. However, closed-system contributory causes — such as the failure of a sensor — may still be important to understand the scenario that led to a violation. Our definition of causation is *counterfactual*, as shown by the next theorem.

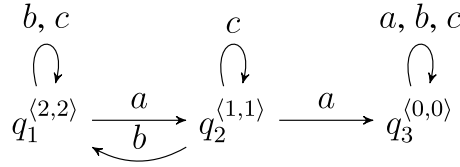


Fig. 8. Semantic graph $[A||O]$ entering q_3 upon two consecutive a 's not separated by a b . The initial state is q_1 . Tuples in superscript denote the level of choice. All transitions are controllable.

Theorem 3. Given a model A , an observer $\mathcal{O}_P$ and a run $r = v_0 \xrightarrow{e_1} v_1 \dots \xrightarrow{e_n} v_n \in \rho(A||\mathcal{O}_P, v_0)$, let ξ be the level of choice of $[A||\mathcal{O}_P]$. Then

1. for any controllable contributory cause $v_{i-1} \xrightarrow{e_i} v_i \in ECT_{\xi_1}(r)$ there exists a counterfactual transition $t' = v_{i-1} \xrightarrow{e'} v' \in \Sigma_c$ such that $\xi_1(v') \geq \xi_1(v_{i-1})$
2. for any closed-system contributory cause $v_{i-1} \xrightarrow{e_i} v_i \in ECT_{\xi_2}(r)$ there exists a counterfactual transition $t' = v_{i-1} \xrightarrow{e'} v' \in \Sigma$ such that $\xi_2(v') \geq \xi_2(v_{i-1})$

In both cases, we call t' a *safe alternative transition*.

Proof. Let $v_{i-1} \xrightarrow{e_i} v_i \in ECT_{\xi_1}(r)$. We have $\xi_1(v_{i-1}) > \xi_1(v_i)$ by definition of $ECT_{\xi_1}(r)$, and therefore with Proposition 2, we have $\xi_1(v_{i-1}) = 1 + \xi_1(v_i) = 1 + \min \{ \xi_1(v'') \mid \exists e'' \in \Sigma_c : v_{i-1} \xrightarrow{e''} v'' \}$ and $|\{ \xi_1(v'') \mid \exists e'' \in \Sigma_c : v_{i-1} \xrightarrow{e''} v'' \}| > 1$ by definition of ξ_1 . Hence $\exists e' \in \Sigma_c : v_{i-1} \xrightarrow{e'} v'$ and $\xi_1(v_{i-1}) \leq \xi_1(v')$. The proof for closed-system contributory causes is analog. $\square$

Safe alternative transitions do not decrease the level of choice. A controllable effective choice transition of $ECT_{\xi_1}(r)$ is therefore indeed a bad choice of the system because it could have been avoided by taking the safe alternative. A closed-system effective choice transition of $ECT_{\xi_2}(r)$ may be uncontrollable and cannot always be avoided via a controllable alternative.

Example 5 (Running example). Let us consider a run r , for example run r_2 given in Table 3, that leads to the transition $t_{21} : v_{19} \xrightarrow{rcr} v_{20}$, of the partial semantic graph of Fig. 6 and ends with the last transition $t_{25} : v_{23} \xrightarrow{meas} v_{24}$.

Transition $t_{21} : v_{19} \xrightarrow{rcr} v_{20}$ is an effective choice transition that belongs to $ECT_{\xi_1}(r_2)$ and to $ECT_{\xi_2}(r_2)$ as it satisfies the conditions of Definition 13, i.e., $\forall i \in [20, 24], \xi_1(v_{19}) > \xi_1(v_i)$ and $\xi_2(v_{19}) > \xi_2(v_i)$. By Theorem 3, there exists at least a safe alternative from its source state v_{19} . This safe alternative is the transition $v_{19} \xrightarrow{rbr} v_{25}$ that is displayed on the bottom level of Fig. 6.

Transition $t_{23} : v_{21} \xrightarrow{fcr} v_{22}$ is also an effective choice transition but it only belongs to $ECT_{\xi_2}(r_2)$ because $\forall i \in [22, 24], \xi_1(v_{21}) > \xi_1(v_i)$ but $\xi_2(v_{21}) \not> \xi_2(v_i)$.

5.5. Construction of choice explanations

Choice explanations take the level of choice into account by retaining effective choice transitions of a given run r that violates safety property P . Moreover, to be fully informative, they must also model what may happen between two effective choice transitions. For instance, if A is an automaton that produces $\{a, b\}^*$, and observer $\mathcal{O}$ enters ko upon two consecutive a 's — the semantic graph $[A||\mathcal{O}]$ is shown in Fig. 8 —, the explanation should be able to express “two consecutive a 's without a b in between”.

For a run r , $ECT_{\xi}(r)$ is the set of transitions that lead effectively the system towards the violation of the safety property P by decreasing definitively ξ_1 or ξ_2 in its suffix. In order to build the explanation, we order the set of effective choice transitions. The order of effective choice transitions, as they happen in r , is total. We define a relaxed order $<$ that does not depend on the observation but only on the level of choice. This order allows for potential concurrency between effective choice transitions and only orders two transitions when it is necessary to ensure their membership in $ECT_{\xi_1}(r)$ and $ECT_{\xi_2}(r)$.

Let us recall some basic definitions. A relation $< \subseteq E \times E$ is a *strict partial order* if it satisfies the conditions of (1) irreflexivity: $\forall e \in E : \neg(e < e)$; (2) transitivity: $\forall e_1, e_2, e_3 \in E : e_1 < e_2 \wedge e_2 < e_3 \implies e_1 < e_3$; and (3) asymmetry: $\forall e_1, e_2 \in E : e_1 < e_2 \implies \neg(e_2 < e_1)$. Given a strict partially ordered set $\langle E, < \rangle$, let $\max(\langle E, < \rangle) = \{ t \in E \mid \forall t' \in E, \neg(t < t') \}$ be the set of maximal elements of E . The union of two relations $\langle E_1, <_1 \rangle$ and $\langle E_2, <_2 \rangle$ is the relation $\langle E_1, <_1 \rangle \cup \langle E_2, <_2 \rangle = \langle E_1 \cup E_2, <_1 \cup <_2 \rangle$.

Choice explanations are hence of the form of a partially ordered set of effective choice transitions, along with a function ϕ that maps ordered tuples in $<$ to what is “in between”. More formally, a choice explanation takes the form

$$\langle \langle E, < \rangle, \phi \rangle$$

where $\langle E, < \rangle$ is a strict partially ordered set (SPO) of effective choice transitions, and ϕ maps ordered tuples in $<$ on an abstraction of the possible behaviours between two effective choice transitions. This abstraction, called *non-choice implicit automaton*, is introduced in Section 5.5.2.

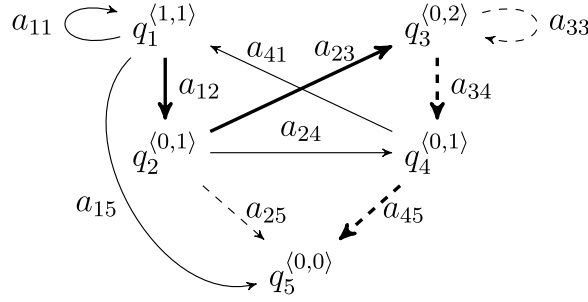


Fig. 9. The initial state is q_1 . Tuples in superscript denote the level of choice. Solid (resp. dashed) transitions are controllable (resp. uncontrollable). The run r is highlighted with thick transitions.

Each contributory cause is encoded in the explanation. Intuitively, a contributory cause corresponds to the occurrence of an event within a context, that leads the system closer to the violation of the safety property. One contributory cause alone may not be sufficient. This is why we encode contributory causes as an ordered set, reflecting concurrency and precedence. Given two contributory causes t_1 and t_2 , $t_1 < t_2$ implies that t_1 happened before t_2 (otherwise t_1 would not be a contributory cause). This order relation allows us to represent exactly the orderings that are causal for the outcome.

Compared to the general form of explanation $\langle \langle E, < \rangle, \phi \rangle$ introduced before, a contrastive choice explanation thus takes the form $\langle \pi, \langle E', <' \rangle, \phi' \rangle$, where π is the contrastive prefix as introduced in Section 4.2. Note that, given a (non-contrastive) explanation $\langle \langle E, < \rangle, \phi \rangle \in \mathcal{E}(\mathcal{A}, \mathcal{O}_P, v_0, tr)$ for some trace $tr = tr_1 \cdot tr_2$ and a contrastive explanation $\langle \pi, \langle E', <' \rangle, \phi' \rangle$ with $\langle \langle E', <' \rangle, \phi' \rangle \in \mathcal{E}(\mathcal{A}, \mathcal{O}_P, v'_0, tr_2)$ for tr , we have $E' \subseteq E$. In words, the contrastive explanation is more concise, as it focuses on explaining only the difference between the expected scenario (the satisfaction of Q) and the violation of P via tr .

5.5.1. Strict partial order of choice

Let us now discuss how to define the partial order $<$ over contributory causes.

Definition 14 (Strict Partial Order of Choice (SPOC)). Given a model $\mathcal{A}$, an observer $\mathcal{O}_P$ and a run $r \in \rho(\mathcal{A} \parallel \mathcal{O}_P, v_0)$, let ξ be the level of choice of $\langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle := [\mathcal{A} \parallel \mathcal{O}_P]$. We define $SPOC_\xi(r)$ as the tuple $\langle ECT_\xi(r), < \rangle$ where:

- $<_1 = \{ \langle v \xrightarrow{e} v', v_1 \xrightarrow{e_1} v'_1 \rangle \in ECT_\xi(r) \times ECT_{\xi_1}(r) \mid \max(\xi_1(v), \xi_1(v')) \geq \xi_1(v_1) \wedge v \xrightarrow{e} v' \neq v_1 \xrightarrow{e_1} v'_1 \}$
- $<_2 = \{ \langle v \xrightarrow{e} v', v_2 \xrightarrow{e_2} v'_2 \rangle \in ECT_\xi(r) \times ECT_{\xi_2}(r) \mid \max(\xi_2(v), \xi_2(v')) \geq \xi_2(v_2) \wedge v \xrightarrow{e} v' \neq v_2 \xrightarrow{e_2} v'_2 \}$
- $<$ is the transitive closure of $(<_1 \cup <_2)$

For a run r of $\mathcal{A} \parallel \mathcal{O}_P$, $SPOC_\xi(r)$ is an ordered set of effective choice transitions of r . The order of $SPOC_\xi(r)$ captures the constraints on the level of choice ξ . For $t, t' \in ECT_\xi(r)$, $t < t'$ implies that if t happens after the last occurrence of t' in some run r' , then either $ECT_{\xi_1}(r) \neq ECT_{\xi_1}(r')$ or $ECT_{\xi_2}(r) \neq ECT_{\xi_2}(r')$. $SPOC_\xi(r)$ is an ordered set of contributory causes for the violation of the safety property by r .

Example 6 (Non-total $SPOC_\xi$).

For the example of Fig. 9 and the run r shown in thick, we have $SPOC_\xi(r) = \{ \langle t_{12}, t_{45} \rangle, \langle t_{34}, t_{45} \rangle \}$ where $t_{12} = q_1 \xrightarrow{a_{12}} q_2$, $t_{34} = q_3 \xrightarrow{a_{34}} q_4$, and $t_{45} = q_4 \xrightarrow{a_{45}} q_5$. Hence, t_{12} and t_{34} are not ordered among each other. The run with effective choice transitions $t_{34}; t_{12}; t_{45}$ has the same SPOC as r .

The next lemma states that the order in $SPOC_\xi$ is consistent with the order of occurrence in r . In other words, $SPOC_\xi$ is a generalisation of r and different runs may have the same $SPOC_\xi$.

Lemma 5. Given a model $\mathcal{A}$ and an observer $\mathcal{O}_P$, let ξ be the level of choice of $\langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle := [\mathcal{A} \parallel \mathcal{O}_P]$. We have

$$\forall r \in \rho(\mathcal{A} \parallel \mathcal{O}_P, v_0) : \langle t, t' \rangle \in SPOC_\xi(r) \implies Last(t, r) < Last(t', r)$$

where

$$Last(v \xrightarrow{e} v', v_0 \xrightarrow{e_1} v_1 \dots \xrightarrow{e_n} v_n) = \max\{i \in [1, n] \mid v_{i-1} \xrightarrow{e_i} v_i = v \xrightarrow{e} v'\}$$

Proof. Let $r \in \rho(\mathcal{A} \parallel \mathcal{O}_P, v_0)$, we use the inductive definition of the transitive closure $<$ of $(<_1 \cup <_2)$, with $<_1, <_2$ defined in Definition 14:

$$<^0 = <_1 \cup <_2$$

$$\forall i \in \mathbb{N}, <^{i+1} = <^i \cup \{ \langle t, t' \rangle \mid \exists t'' \in E, \langle t, t' \rangle \in <^i \wedge \langle t', t'' \rangle \in <^i \}$$

We prove by induction on $i \in \mathbb{N}$ that $\langle t, t' \rangle \in <^i \implies Last(t, r) < Last(t', r)$

For $<^0 = <_1 \cup <_2$, $t, t' \in ECT_\xi(r)$, if we have $t <_1 t'$ (resp. $t <_2 t'$) then any occurrence of t after the last occurrence of t' implies that $t' \notin ECT_{\xi_1}(r)$ (resp. $t' \notin ECT_{\xi_2}(r)$), hence the property is satisfied.

For the induction step, if we have $t \prec^{i+1} t' \wedge \exists t'' \in ECT_{\xi}(r), t \prec^i t'' \wedge t'' \prec^i t'$ we have :

$t'' \prec^i t'$ implies, with the induction hypothesis, that there is no occurrence of t'' after the last occurrence of t' and therefore the last occurrence of t'' happens before t' .

$t \prec^i t''$ implies, with the induction hypothesis, that there is no occurrence of t after the last occurrence of t'' , hence there is no occurrence of t after the last occurrence of t' . $\square$

Proposition 3. Given a model $\mathcal{A}$ and an observer $\mathcal{O}_p$, let ξ be the level of choice of $\langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle := [\mathcal{A} \parallel \mathcal{O}_p]$. For all runs $r \in \rho(\mathcal{A} \parallel \mathcal{O}_p, v_0)$, the relation $SPOC_{\xi}(r)$ is an SPO.

Proof. $\forall t_1, t_2 \in E$, such that $t_1 < t_2$, we have $t_1 < t_2 \implies Last(t_1, r) < Last(t_2, r)$ (Lemma 5). Therefore, $t_1 < t_2 \implies \neg(t_2 < t_1)$.

Irreflexivity is a subcase of the proof above with $t_1 = t_2$.

Finally, in Definition 14, $<$ is the transitive closure of $(\prec_1 \cup \prec_2)$ and is therefore transitive. $\square$

The relation $SPOC_{\xi}(r)$ being an SPO, it has the properties that have been shown desirable to encode the pattern of contributory causes.

The next lemma states that if a run r violates the safety property then its last observed effective choice transition is maximal in $SPOC_{\xi}$ and leads to $\xi = \langle 0, 0 \rangle$.

Lemma 6. Given a model $\mathcal{A}$ and an observer $\mathcal{O}_p$, let ξ be the level of choice of $\langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle := [\mathcal{A} \parallel \mathcal{O}_p]$, $\forall r \in \rho^F(\mathcal{A} \parallel \mathcal{O}_p, v_0)$ such that $ECT_{\xi}(r) \neq \emptyset$, $SPOC_{\xi}(r)$ has a unique maximal element $v \xrightarrow{e} v'$ such that $\xi(v') = \langle 0, 0 \rangle$.

Proof. Let $r = v_0 \xrightarrow{e_1} v_1 \dots v_{n-1} \xrightarrow{e_n} v_n \in \rho^F(\mathcal{A} \parallel \mathcal{O}_p, v_0)$ and $i = \min\{j \in [0, n] \mid \xi(v_j) = \langle 0, 0 \rangle\}$.

i is well defined because $r \in \rho^F(\mathcal{A} \parallel \mathcal{O}_p, v_0) \implies v_n \in \neg P \implies \xi(v_n) = \langle 0, 0 \rangle$

Furthermore $ECT_{\xi}(r) \neq \emptyset$ and i minimal implies $\xi(v_{i-1}) > \langle 0, 0 \rangle$.

Proposition 2 implies that $\xi(v_{i-1})$ is either $\langle 0, 1 \rangle$ or $\langle 1, 1 \rangle$, and the set of states where $\xi_2 = 0$ is closed under the transition relation, hence $v_{i-1} \xrightarrow{e_i} v_i \in ECT_{\xi_2}(r)$.

$v_{i-1} \xrightarrow{e_i} v_i$ is maximal w.r.t. $\prec_2$ because $\forall v \in \mathcal{V}, \xi_2(v) \geq 0 = \xi_2(v_i)$.

$v_{i-1} \xrightarrow{e_i} v_i$ is maximal w.r.t. $\prec_1 \cup \prec_2$ because either it is also maximal in $prec_1$ whenever $v_{i-1} \xrightarrow{e_i} v_i \in ECT_{\xi_1}(r)$ or it is not in the relation $\prec_2$.

To conclude, the transitive closure does not change the maximal element of the relation and $v_{i-1} \xrightarrow{e_i} v_i \in ECT_{\xi_1}(r)$ is the unique maximal element in $SPOC_{\xi}(r)$. $\square$

Lemma 6 is crucial because it states that from a state v' such that $\xi(v') = \langle 0, 0 \rangle$, a violation of the safety property observed by $\mathcal{O}_p$ is inevitable. The maximal effective choice transition in $SPOC_{\xi}(r)$ is the last effective choice transition happening in r (Lemma 5).

Example 7 (Running example). Consider again the run r_2 of $[\mathcal{A} \parallel \mathcal{O}_p]$ starting in v^0 and composed of 25 transitions as well as its corresponding trace tr_2 as detailed in Table 3. The contrastive choice explanation of the violation of $P = sd > 0$ in contrast to keeping $Q = sd \geq 3$ in tr_2 takes the form $\langle \pi, \langle E', < \rangle, \phi \rangle$. The determination of π and E is outlined below, while ϕ will be determined in the following example.

The maximal sequence in tr_2 that belongs to $safeLang_{\mathcal{A}}(\mathcal{O}_Q)$ is the single event fbr , hence $\pi = fbr.ra$ and $\rho_{\mathcal{A}}(\pi) = v_2$, which is the source state of $t_3(v_2 \xrightarrow{meas} v_3)$. The safety property Q is then violated as soon as the distance is updated to $sd = 2$ with $t_3(v_2 \xrightarrow{meas} v_3)$. Note that applying a cruise event rcr instead of ra would safely keep a distance of 3 with the front car.

The contrastive suffix $r_c = r_2 \setminus r_{\pi}$ corresponds to the trace $t_3 \dots t_{25}$, where $r_{\pi} = v_0 \xrightarrow{fbr} v_1 \xrightarrow{ra} v_2$. The set of effective choice transitions is as follows:

- $ECT_{\xi_1}(r_c)$ is $\{t_{21}\}$: in this scenario, the rear car makes one impacting choice that is rcr at $t_{21}(v_{19} \xrightarrow{rcr} v_{20})$. This choice decreases the controllable level of choice ξ_1 from level 1 (initial level) to level 0 and is hence a contributory cause. After t_{21} , the violation of P only depends on the behaviour of the front car. Instead of cruising (rcr), the rear car should have break (rbr). Indeed the transition $v_{19} \xrightarrow{rbr} v_{25}$ is a safe alternative to $t_{21}(v_{19} \xrightarrow{rcr} v_{20})$, as illustrated in Fig. 6).
- $ECT_{\xi_2}(r_c)$ is $\{t_{11}, t_{12}, t_{14}, t_{17}, t_{20}, t_{23}\}$. The first impacting choice, hence first contributory cause, is made by the rear car at $t_{11}(v_9 \xrightarrow{ra} v_{10})$ that starts accelerating while the front car is at a distance of 5m, which reduces the number of impacting system-wide choices that can lead to the violation of P . The second uncontrollable effective choice transition is the occurrence of the failure at $t_{12}(v_{10} \xrightarrow{fail} v_{11})$ of the sensor that leads the rear car to read an erroneous distance between both cars and to accelerate a second time at $t_{15}(v_{13} \xrightarrow{ra} v_{14})$ (uncontrollable level of choice changes from 6 to 5). Remark that t_{15} is not considered as an effective choice transition: based on the measured distance, the rear car must indeed accelerate, it is a direct consequence of the occurrence of the failure. The second and third uncontrollable events are the break events of $t_{14}(v_{12} \xrightarrow{fbr} v_{13})$ and $t_{17}(v_{15} \xrightarrow{fbr} v_{16})$ of the front car, leading to the uncontrollable level of choice $\xi_2 = 3$. Finally, as the front car keeps stopped, i.e. $fcs = 0$, after t_{17} , the system reaches inevitable states and finally violates P .

To summarise:

- $\pi = fbr.ra$
- $r_\pi = v_0 \xrightarrow{fbr} v_1 \xrightarrow{ra} v_2$
- $r_c = t_3 \dots t_{25}$
- $E' = SPOC_\xi(r_c = r_2 \setminus r_\pi) = \{t_{11} < t_{12} < t_{14} < t_{17} < t_{20} < t_{21} < t_{23}\}$

5.5.2. Non-choice implicit automaton

States whose level of choice is increased with respect to their successors are qualified as *choice states* and have led to the definition of effective choice transitions. *Non-choice states* are the counterpart of choice states. When an event occurs in a non-choice state it is not relevant to be part of the explanation. However non-choice states are important to characterise what happens in between two effective choice transitions, which will be captured by a *non-choice implicit automaton* as defined subsequently.

The following definition defines, for any transition t , specific upper bounds $\bar{\xi}_1^{(r)}(t)$ (resp. $\bar{\xi}_2^{(r)}(t)$) on ξ_1 (resp. ξ_2).

Definition 15 (Upper bounds on ξ). Given a model $\mathcal{A}$, an observer $\mathcal{O}_P$ and a run $r \in \rho(\mathcal{A} \parallel \mathcal{O}_P, v_0)$, let ξ be the level of choice of $\langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle := [\mathcal{A} \parallel \mathcal{O}_P]$ and let $\langle ECT_\xi(r), < \rangle = SPOC_\xi(r)$. We define $\bar{\xi}_1^{(r)}, \bar{\xi}_2^{(r)} : (\rightarrow) \rightarrow (\mathbb{N} \cup \{\infty\})$ such that:

- $\forall t \in ECT_\xi(r)$:

$$\bar{\xi}_1^{(r)}(t) = \min\{\xi_1(v') \mid v \xrightarrow{e} v' \in pre(t) \cap ECT_{\xi_1}(r)\}$$

$$\bar{\xi}_2^{(r)}(t) = \min\{\xi_2(v') \mid v \xrightarrow{e} v' \in pre(t) \cap ECT_{\xi_2}(r)\}$$

where $pre(t) = \{t' \in \rightarrow \mid t' < t\} \cup \{t\}$ and $\min \emptyset = \infty$;

- $\forall t \in (\rightarrow \setminus ECT_\xi(r)) : \bar{\xi}_1^{(r)}(t) = \infty$ and $\bar{\xi}_2^{(r)}(t) = \infty$

For an effective choice transition t , the upper bounds $\bar{\xi}_1^{(r)}(t)$ and $\bar{\xi}_2^{(r)}(t)$ are the maximal values of ξ_1 and ξ_2 such that all the effective choice transitions smaller or equal to t in $SPOC_\xi(r)$ remain effective in all runs containing only states v such that $\xi_1(v) \leq \bar{\xi}_1^{(r)}(t) \wedge \xi_2(v) \leq \bar{\xi}_2^{(r)}(t)$ in the suffix after the last occurrence of t . We call $\bar{\xi}_1^{(r)}(t)$ and $\bar{\xi}_2^{(r)}(t)$ upper bounds because they are upper bounds on ξ_1 and ξ_2 for all the states visited after t in r . More generally, for any alternative run r' that shares with r the same prefix before the last occurrence of t (the effective one), if the levels of choice ξ_1 and ξ_2 of the states that follow t are not lower or equal to the upper bounds $\bar{\xi}_1^{(r)}(t)$ or $\bar{\xi}_2^{(r)}(t)$, then either t or a previous transition $t' \in ECT_\xi(r)$ is no longer effective in the run r' . Upper bounds are defined for any transition t but if $t \notin ECT_\xi(r)$ then both upper bounds are defined as infinite.

For $t, t' \in ECT_\xi(r)$ such that $t < t'$ and there is no transition $t'' \in ECT_\xi(r)$ such that $t < t'' \wedge t'' < t'$, we say that (t, t') is *consecutive*.

We can now define the *non-choice states* and the non-choice implicit automaton that characterises all the runs between consecutive effective choice transitions which maintain the effective choice transitions as in the explained run r .

Definition 16 (Non-choice states). Given a model $\mathcal{A}$ and an observer $\mathcal{O}_P$, let ξ be the level of choice of $\langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle := [\mathcal{A} \parallel \mathcal{O}_P]$. For any $r \in \rho^F(\mathcal{A} \parallel \mathcal{O}_P, v_0)$ and any pair $v_1 \xrightarrow{e_1} v'_1, v_2 \xrightarrow{e_2} v'_2$ of consecutive effective choice transitions in $SPOC_\xi(r)$, the set of *non-choice states* $NCS'(v_1 \xrightarrow{e_1} v'_1, v_2 \xrightarrow{e_2} v'_2)$ is the maximal subset $\mathcal{V}'$ of $\mathcal{V}$ such that:

- $\forall v \in \mathcal{V}' (\xi_1(v) \leq \bar{\xi}_1^{(r)}(v_1 \xrightarrow{e_1} v'_1) \wedge (\xi_2(v) \leq \bar{\xi}_2^{(r)}(v_2 \xrightarrow{e_2} v'_2)))$, and
- from any $v \in \mathcal{V}'$ there is a run to v_2 in $\rightarrow \cap (\mathcal{V}' \times \Sigma \times \mathcal{V}')$

The restriction $[\mathcal{A} \parallel \mathcal{O}_P] \Big|_{NCS'(v_1 \xrightarrow{e_1} v'_1, v_2 \xrightarrow{e_2} v'_2)}$ is the LTS

$$\langle \mathcal{V}', \Sigma, \rightarrow \cap (\mathcal{V}' \times \Sigma \times \mathcal{V}'), v'_1, \{v'_2\} \rangle.$$

When $v_1 \xrightarrow{e_1} v'_1, v_2 \xrightarrow{e_2} v'_2$ are two consecutive effective choice transitions in $SPOC_\xi(r)$, the LTS $[\mathcal{A} \parallel \mathcal{O}_P] \Big|_{NCS'(v_1 \xrightarrow{e_1} v'_1, v_2 \xrightarrow{e_2} v'_2)}$ denotes

all the potential suffixes of runs after $v_1 \xrightarrow{e_1} v'_1$ such that:

- all the effective choice transitions lower or equal to $v_1 \xrightarrow{e_1} v'_1$ in $SPOC_\xi(r)$ are effective choice transitions,
- the origin v_2 of the next effective choice transition $v_2 \xrightarrow{e_2} v'_2$ is reached.

Example 8 (Non-total $SPOC_\xi$ continued). For the effective choice transitions of the example of Fig. 9, we obtain the upper bounds

$$\bar{\xi}_1^{(r)}(t_{12}) = \bar{\xi}_1^{(r)}(t_{34}) = \bar{\xi}_1^{(r)}(t_{45}) = \bar{\xi}_2^{(r)}(t_{45}) = 0$$

$$\bar{\xi}_2^{(r)}(t_{12}) = \bar{\xi}_2^{(r)}(t_{34}) = 1$$

and therefore the non-choice states

$$NCS'(t_{12}, t_{45}) = NCS'(t_{34}, t_{45}) = \{q_2, q_4\}$$

The next definition will be used to encode the semantics of choice explanations in [Section 5.6](#). For a choice explanation of a trace tr , we want to define all the traces tr' that have the same effective choice transitions.

The effectiveness of choice transitions is maintained as long as the segments of runs between two consecutive effective choice transitions remain within non-choice states, as defined in [Definition 16](#). That is, these segments do not contain any safe alternatives that increase the level of choice, which would otherwise compromise the effectiveness of the preceding choice transitions.

Since the semantics of an explanation are to be computed without the observer $\mathcal{O}_P$ and without the level of choice ξ , the next definition extracts the part $\phi(t_1, t_2)$ of the observer $\mathcal{O}_P$ required to encode the constraints on the level of choice in the non-choice states between two consecutive effective choices.

Definition 17 (Non-choice implicit automaton). Given a model $\mathcal{A}$ with variables X , an observer $\mathcal{O}_P = \langle \Sigma_{\mathcal{O}_P}, X_{\mathcal{O}_P}, \text{inv}_{\mathcal{O}_P}, \delta_{\mathcal{O}_P}, \nu_{\mathcal{O}_P}^0, F_{\mathcal{O}_P} \rangle$, and a run $r \in \rho^F[\mathcal{A} \parallel \mathcal{O}_P, v_0]$, let $\langle \mathcal{V}, \Sigma, \rightarrow, \nu^0, \mathcal{V}^F \rangle = [\mathcal{A} \parallel \mathcal{O}_P]$. For all consecutive pairs $t_1 = v_1 \xrightarrow{e_1} v'_1$, $t_2 = v_2 \xrightarrow{e_2} v'_2$ of effective choice transitions, the non-choice implicit automaton $\phi(t_1, t_2)$ is the implicit automaton $\langle \Sigma_{\mathcal{O}_P}, X \cup X_{\mathcal{O}_P}, \text{inv}, \delta_{\mathcal{O}_P}, \nu_1^0, F_2 \rangle$ where

- $\text{inv} = \bigvee_{v \in \text{NCS}^r(t_1, t_2)} v$
- $F_2 = \bigwedge_{x \in X} (x = v_2(x))$.

Non-choice implicit automata do not contain any contributory causes nor safe alternatives. Rather, we use them to provide context in the explanation: they explain, in a run r encompassing more than one contributory cause, i.e. $|\text{ECT}_{\xi}(r)| \geq 2$, the behaviour that led from one cause to the next in r , as well as the set of equivalent behaviours that do not alter the status of contributory cause to the transitions of $\text{ECT}_{\xi}(r)$.

In order to build a concise explanation, non-choice implicit automata extract only the necessary part of the observer with respect to the effective choice transitions of the explained violation.

Example 9 (Non-total SPOC_{ξ} continued). For our example of [Fig. 9](#), the non-choice implicit automaton is the sub-automaton with the single transition $q_2 \xrightarrow{a_{24}} q_4$.

Example 10 (Running example). Considering the running example, to complete the choice explanation $\langle \pi, \langle E', < \rangle, \phi \rangle$ of the violation of $P = sd > 0$ in contrast to keeping $Q = sd \geq 3$ in the trace tr_2 corresponding to run r_2 detailed in [Table 3](#), one needs to define the non-choice implicit automaton for every pair of consecutive effective choice transitions of $E' = \text{SPOC}_{\xi}(r_c) = \{t_{11} < t_{12} < t_{14} < t_{17} < t_{20} < t_{21} < t_{23}\}$. These automata have different state invariants but they are all defined on the same alphabet, variables and transition function specified below:

- their alphabet $\Sigma_{\mathcal{O}_P} = \{\text{meas}\}$,
- their set of variables is composed of the variables of the model $FC \parallel CC \parallel SR$ detailed in [Table 2](#) and the variable p_{ko} from the observer $\mathcal{O}_P$,
- their transition function is the transition function of the observer $\mathcal{O}_P$:

$$\delta_P(\text{meas}) = (\text{true}, \{ p_{ko} \leftarrow \text{true}, \text{ if } (\neg p_{ko} \wedge sd + rcs - fcs = 0) \})$$

To illustrate the process of determining non-choice states, let us consider the couple of consecutive effective choice transitions $\langle t_{17}, t_{20} \rangle$. The set of non-choice states $\text{NCS}^r(t_{17}, t_{20})$ gathers all the states of $[\mathcal{A} \parallel \mathcal{O}_P]$ in path from the target state of t_{17} , i.e., v_{15} , to the source state of t_{20} , i.e., v_{18} , such that their level of choice is lower or equal to the upper bounds $\xi_1^{(r)}(t_{17}) = \infty$ and $\xi_2^{(r)}(t_{17}) = 3$ (to maintain effective all the effective choice transitions smaller or equal to t_{17} in $\text{SPOC}_{\xi}(r_c)$, where $r_c = r_2 \setminus r_{\pi}$ and r_{π} has been determined as $r_{\pi} = v_0 \xrightarrow{\text{fbr}} v_1 \xrightarrow{\text{ra}} v_2$). $\text{NCS}^r(t_{17}, t_{20})$ can be given as follows:

$$\begin{aligned} \text{NCS}^r(t_{17}, t_{20}) &= rcs = 2 \wedge fcs = 0 \wedge sfm = 0 \wedge \\ &((smd = 10 \wedge sd = 5 \wedge cm \in \{1, 2\}) \\ &\vee (smd = 6 \wedge sd = 3 \wedge cm = 0)) \end{aligned}$$

[Fig. 10](#) graphically represents the three non-choice states in $\text{NCS}^r(t_{17}, t_{20})$. Note that the state v_{27} , target of the event fa , is also involved in a path from t_{17} to t_{20} but its level of choice is greater than $\xi_2^{(r)}(t_{17}) = 3$, so it is not part of $\text{NCS}^r(t_{17}, t_{20})$ as well as all of its successors along the paths leading to v_{18} represented by the dotted arrow in the figure.

[Lemma 7](#) makes the connection between the computation of explanations that depends on the observer $\mathcal{O}_P$ and the computation of the semantics that does not require the observer.

Lemma 7 (Lemma of connection). Given a behavioural model $\mathcal{A}$, an observer $\mathcal{O}_P$ and a run $r \in \rho^F(\mathcal{A} \parallel \mathcal{O}_P, v_0)$, let ξ be the level of choice of $\langle \mathcal{V}, \Sigma, \rightarrow, \nu^0, \mathcal{V}^F \rangle := [\mathcal{A} \parallel \mathcal{O}_P]$. For any pair of effective choice transitions $\langle t_1, t_2 \rangle \in \text{SPOC}_{\xi}(r)$, we have $\rho^F(\mathcal{A} \parallel \mathcal{O}_P) \Big|_{\text{NCS}^r(t_1, t_2)} = \rho^F(\mathcal{A}_{\nu} \parallel \phi(t_1, t_2))$ where ν is the target state of t_1 .

Proof. Let us define the two semantic LTS

$$G = [\mathcal{A} \parallel \mathcal{O}_P] \Big|_{\text{NCS}^r(t_1, t_2)} = \langle \mathcal{V}_G, \Sigma_G, X_G, \rightarrow_G, \nu_G^0, \mathcal{V}_G^F \rangle$$

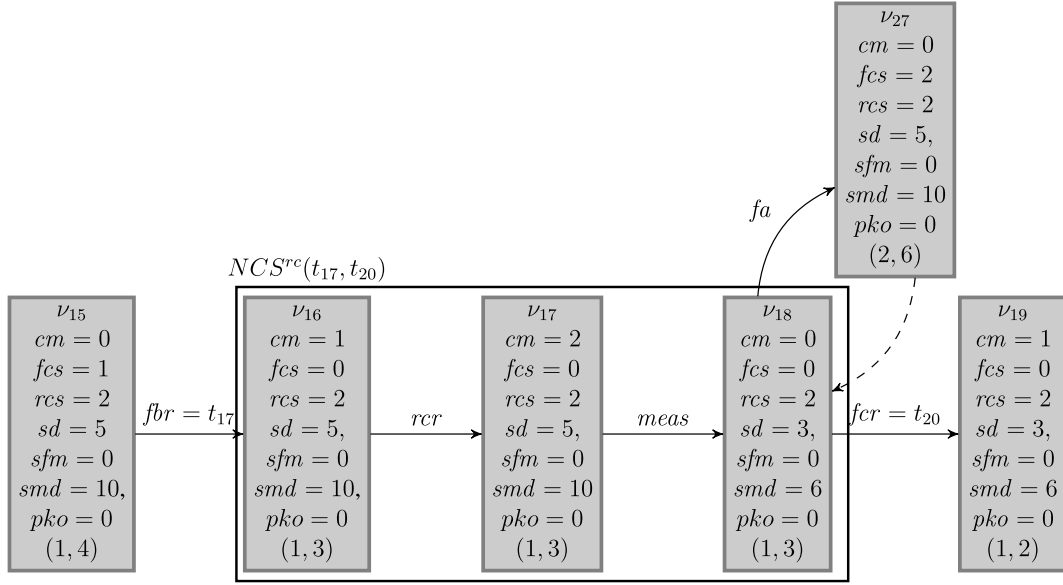


Fig. 10. Part of $[A \parallel \mathcal{O}_P]$ where the big rectangle gathers the states of $NCS^{rc}(t_{17}, t_{20})$.

$$G' = [\mathcal{A}_{v'_1} \parallel \phi(t_1, t_2)] = \langle \mathcal{V}_{G'}, \Sigma_{G'}, X_{G'}, \rightarrow_{G'}, v_{G'}^0, \mathcal{V}_{G'}^F \rangle$$

and the states v_1, v'_1, v_2, v'_2 and events e_1, e_2 such that $t_1 = v_1 \xrightarrow{e_1} v'_1$ and $t_2 = v_2 \xrightarrow{e_2} v'_2$.

Firstly, we prove that $\mathcal{V}_G = \mathcal{V}_{G'} = NCS^r(t_1, t_2)$. By definition of the composition $\mathcal{A}_{v'_1} \parallel \phi(t_1, t_2)$ and the semantic LTS G' , a state $v \in \mathcal{V}_{G'}$ satisfies the invariant of $\phi(t_1, t_2)$, and therefore is a state of $NCS^r(t_1, t_2)$ (cf Definition 17). Inversely, a state in $NCS^r(t_1, t_2)$ is in $\mathcal{V}_{G'}$ by Definition 17 of $\phi(t_1, t_2)$. Hence, the states of G and G' are exactly $NCS^r(t_1, t_2)$.

Secondly, we have $\delta_p = \delta_{\mathcal{O}_P}$ (Definition 17), therefore the syntactic transitions of $[\mathcal{A}_{v'_1} \parallel \phi(t_1, t_2)]$ are syntactic transitions of $[A \parallel \mathcal{O}_P]$. Hence, the reachable transitions in G and G' from the state v'_1 are the same.

Finally, the accepting states of G are $\{v_2\}$ and the initial state of G is v'_1 and all the runs in $\rho^F(\mathcal{A}_{v'_1} \parallel \phi(t_1, t_2))$ start in v'_1 and end in the state v_2 (Definition 3). $\square$

5.6. Choice explanations and their semantics

Intuitively, the semantics of a causal explanation for the violation of a safety property is the set of behaviours that share the causes that are reported in the explanation. Hence, the semantics of an explanation is a language of the behaviours having the same effective choice transitions and respecting the same SPOC as the run to be explained.

Interestingly, one can compute the semantics of an explanation without knowing the violated safety property or the respective observer. The motivation for this constraint is to avoid the potential complexity of the observer. It enables explanations to be compared with one another through their underlying semantics.

Definition 18 (Choice Explanation).

Given a behavioural model $\mathcal{A}$, an observer $\mathcal{O}_P$ of the safety property P , let ξ be the level of choice function of $\langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle := [A \parallel \mathcal{O}_P]$, for all runs $r = v_0 \xrightarrow{e_1} \dots \xrightarrow{e_n} v_n \in \rho^F(\mathcal{A} \parallel \mathcal{O}_P, v_0)$ violating P , the *choice explanation (CE)* of the trace $e_1 \dots e_n$ is

$$CE(\mathcal{A}, \mathcal{O}_P, v_0, e_1 \dots e_n) = \langle SPOC_{\xi}(r), \phi \rangle$$

where ϕ defines the non-choice implicit automata between consecutive transitions of $SPOC_{\xi}(r)$, according to Definition 17.

CE returns a unique explanation for a given trace. The first part of the explanation is the partially ordered set $SPOC_{\xi}(r)$ of effective choice transitions. The maximal element is the contributory cause of r that leads to a state where the violation of the safety property P is inevitable. The rest of the elements is the set of contributory causes in r that lead to the violation of P . $SPOC_{\xi}(r)$ encodes the occurrences of the contributory causes in the explanation.

The second element of the explanation is a function ϕ that encodes an invariant for each couple of consecutive contributory causes with an implicit automaton. The invariants give the context in which the contributory causes are effective.

With the definition of choice explanations, we define their semantics. As opposed to the construction of a choice explanation, semantics only depends on the explanation and the behavioural model but, as already said, not on the observer.

Definition 19 (Semantics of choice explanations). Given a model $\mathcal{A}$ such that $[\mathcal{A}] = \langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle$, an *explanation* $\varepsilon = \langle \langle E, < \rangle, \phi \rangle$, and a state $v_0 \in \mathcal{V}$, the semantics $\text{sem}_{\mathcal{A}}(\varepsilon, v_0)$ of ε from v_0 is

$$\text{sem}_{\mathcal{A}}(\varepsilon) = \{tr(r) \mid r \in \rho(\mathcal{A}, v_0) \wedge \text{map}(\varepsilon, r)\}$$

where $\text{map}(\varepsilon, v_0 \xrightarrow{e_1} v_1 \xrightarrow{e_2} \dots \xrightarrow{e_n} v_n) = \exists \psi : E \rightarrow [1, n]$ injective and monotonic w.r.t. $<$ such that

1. $\forall t \in E : t = v_{\psi(t)-1} \xrightarrow{e_{\psi(t)}} v_{\psi(t)}$ and
2. $\forall \langle t_1, t_2 \rangle \in \text{dom}(\phi) : e_{\psi(t_1)+1}, \dots, e_{\psi(t_2)-1} \in L^F(\mathcal{A} \mid v'_1 \parallel \phi(\langle t_1, t_2 \rangle))$ where v'_1 is the target state of t_1 .

A trace w in the semantics of an explanation $\langle \langle E, < \rangle, \phi \rangle$ is the trace of some run of the model $\mathcal{A}$ such that, from item 1. of Definition 19, an occurrence function ψ monotonically maps each transition in E to a transition of r , and, from item 2. of Definition 19, for each pair $\langle t_1, t_2 \rangle$ of transitions in ϕ , the states of r between transition t_1 and t_2 satisfy the invariant $\phi(t_1, t_2)$.

Example 11. For the example of Fig. 8 and the trace $a \cdot c \cdot a$, we have

$$\text{SPOC}_{\varepsilon}(r) = \{t_1 < t_2\}$$

with $t_1 = q_1 \xrightarrow{a} q_2, t_2 = q_2 \xrightarrow{a} q_3$, and r the run of tr . The upper bounds are

$$\bar{\xi}_1^{(r)}(t_1) = \bar{\xi}_2^{(r)}(t_1) = 1$$

$$\bar{\xi}_1^{(r)}(t_2) = \bar{\xi}_2^{(r)}(t_2) = 0$$

which gives us the non-choice states $\text{NCS}^r(t_1, t_2) = \{q_2\}$. The non-choice implicit automaton $\phi(t_1, t_2)$ models the semantics

$$\begin{array}{c} c \\ \downarrow \\ q_2 \end{array}$$

That is, the choice explanation $\varepsilon = \langle \text{SPOC}_{\varepsilon}(r), \phi \rangle$ indeed expresses the fact that the safety violation is caused by two consecutive a 's possibly separated only by c 's. The semantics is

$$\text{sem}_{\mathcal{A}}(\varepsilon) = (b + c + ab)^* \cdot \mathbf{a} \cdot c^* \cdot \mathbf{a} \cdot (a + b + c)^*$$

where labels of the effective choice transitions t_1 and t_2 are printed in bold.

Example 12 (Non-total $\text{SPOC}_{\varepsilon}$ continued). For the explanation $\varepsilon = \langle \langle E, < \rangle, \phi \rangle$ of Example 6 and Fig. 9, we have

$$\text{sem}_{\mathcal{A}}(\varepsilon) = L^F(\mathcal{A}) \cap (\Sigma^* \cdot (\mathbf{a}_{12} \cdot a_{23} \cdot \mathbf{a}_{33}^* \cdot \mathbf{a}_{34} \cdot \mathbf{a}_{45} + \mathbf{a}_{34} \cdot a_{41} \cdot \mathbf{a}_{11}^* \cdot \mathbf{a}_{12} \cdot a_{24} \cdot \mathbf{a}_{45}))$$

where labels of effective choice transitions are printed in bold. The semantics thus consists of all traces that for which the effective choice transitions are those of the explanation, respecting $<$. Notice that ε characterises exactly the traces that subsequently decrease their controllable and closed-system levels of choice through events a_{12} and a_{34} , in any order, followed by event a_{45} entering the violating state q_5 . The explanation is thus an abstract representation of the run shown in Fig. 9 that preserves the causally pertinent information that led to the failure, while abstracting away irrelevant information (the order of a_{12} and a_{34} , and events that do not alter the level of choice, taking place between the contributory causes).

In the next subsection, we prove the expected properties of explanations defined in Section 4 for choice explanations. We also prove another property (Theorem 4) that justifies the intuition we gave on the semantics with respect to the causes of a trace. The contributory causes of the explained trace are also contributory causes of all traces of the semantics of the choice explanation. That causal property implies the satisfaction of the expected properties on choice explanations.

Example 13 (Running example). Considering again the choice explanation ε detailed in Examples 7 and 10 for run r_2 , the semantics $\text{sem}_{\mathcal{A}}(\varepsilon)$ then represents any possible trace $tr = tr_1 \cdot tr_2$ from $[\mathcal{A}][\mathcal{O}_P]$ defined as follows.

1. Trace tr_1 is any trace associated with a run from $[\mathcal{A}][\mathcal{O}_P]$ starting from state $v_2 = (2, 0, 1, 3, 1, 3, 0)$ leading to the state $v_9 = (1, 2, 0, 5, 1, 5, 0)$ (see Table 3). Note that by construction trace tr_1 does not violate P .
2. tr_2 is a trace that belongs to the suffix language of trace tr_1 in the language $\text{sem}_{\mathcal{A}}(\varepsilon)$ that corresponds to the following regular expression

$$\begin{array}{l} \mathbf{ra.fail.meas.fbr.ra.meas.fbr.rcr.meas.fcr.rcr.} \\ \mathbf{meas.fcr.rbr.(meas + (meas.(fcr + fa)))} \end{array}$$

Events in bold here are associated with the effective choice transitions in the explanation ε .

The overall semantics $\text{sem}_{\mathcal{A}}(\varepsilon)$ is a language that is represented by a minimal deterministic automaton consisting of 246 states and 415 transitions while the associated choice explanation ε consists of only 7 transitions that implicitly characterise its semantics.

5.7. Choice explanation properties

The following theorem establishes the correspondence between the contributory causes of a trace and the semantics of a choice explanation. It states that a trace is in the semantics of a choice explanation of tr if and only if its contributory causes (with respect to the safety property P) contain the contributory causes of tr . This theorem is important because all the other theorems on the semantics of explanations are corollaries.

Theorem 4. *Given a behavioural model $\mathcal{A}$, an observer $\mathcal{O}_P$ of the safety property P , a state v_0 of $\mathcal{A} \parallel \mathcal{O}_P$, and a run $r \in \rho^F(\mathcal{A} \parallel \mathcal{O}_P, v_0)$, let ξ be the level of choice of $[\mathcal{A} \parallel \mathcal{O}_P]$, and $\langle\langle E, < \rangle, \phi \rangle = CE(\mathcal{A}, \mathcal{O}_P, v_0, tr(r))$. For all runs $r' \in \rho(\mathcal{A} \parallel \mathcal{O}_P, v_0)$,*

$$tr(r') \in sem_{\mathcal{A}}(\langle\langle E, < \rangle, \phi \rangle) \iff ECT_{\xi_1}(r) \subseteq ECT_{\xi_1}(r') \wedge ECT_{\xi_2}(r) \subseteq ECT_{\xi_2}(r').$$

For sake of notational simplicity, let us denote $tr = tr(r)$ and $tr' = tr(r')$. We prove each implication separately. For the first implication from left to right, we prove that the explanation encodes the effective choice transitions properly. For the second implication from right to left, we have to prove that, when a trace tr' has more effective choice transitions than tr , the stronger causal constraints in tr' imply the requirements on the semantics of the explanation of tr .

Proof. Let $r' = v_0 \xrightarrow{e_1} v_1 \dots \xrightarrow{e_n} v_n \in \rho(\mathcal{A} \parallel \mathcal{O}_P, v_0)$ and $tr' := tr(r')$.

Implication from left to right. Suppose that $tr' \in sem_{\mathcal{A}}(\langle\langle E, < \rangle, \phi \rangle)$. There exists an occurrence function ψ of tr' as defined in [Definition 19](#) because $tr' \in sem_{\mathcal{A}}(\langle\langle E, < \rangle, \phi \rangle)$. Let $i = \{\psi(t) \mid t \in E\}$ the set of indices on which the transitions in E are mapped. We prove by induction on $i \in i$ that (IH) all transitions $t \in E$ with $\psi(t) \geq i$ are effective choice in r' , and $\forall j \in [\psi(t), n] : \xi(v_{\psi(t)-1}) > \xi(v_j)$.

First, $t \in E$ with $i := \psi(t) = \max i$ is, by [Lemma 6](#), the unique maximal element of $SPOC_{\xi}(r)$ and $\xi(v_i) = \langle 0, 0 \rangle$, hence, by construction of ξ , $\xi(v_j) = \langle 0, 0 \rangle$ for $i \leq j \leq n$. Since i is an (effective) choice transition in r , $\xi(v_{i-1}) > \xi(v_i)$. Therefore t is effective choice in r' .

Now suppose that, for some $i \in i$ with $\min i < i$, all transitions $t \in E$ with $i \leq \psi(t)$ are effective choice in r' . Let $t = \psi(i)$, $i' = \max\{j \in i \mid j < i\}$, and $t' \in E$ be such that $\psi(t') = i'$. By [Definitions 16, 17, and 19](#), the non-choice automaton $\phi(t', t)$ bounds the level of choice of the states visited by r' between t' and t such that $\xi(v_{i'}) \geq \xi(v_j)$ for $i' \leq j < i$. By induction hypothesis, $\xi(v_{i-1}) > \xi(v_j)$ for $i \leq j \leq n$. Hence, (IH) holds for i' , and t' is effective choice also in r' .

Implication from right to left. Let us suppose that

$$ECT_{\xi_1}(r) \subseteq ECT_{\xi_1}(r') \wedge ECT_{\xi_2}(r) \subseteq ECT_{\xi_2}(r'). \quad (9)$$

We define the occurrence function $\psi : E \rightarrow [1, n]$ such that $\forall v \xrightarrow{e} v' \in ECT_{\xi}(r)$, $\psi(v \xrightarrow{e} v')$ is the last occurrence of $v \xrightarrow{e} v'$ in r' (by hypothesis, there is at least one occurrence). We prove that tr' and ψ satisfy the requirements of $sem_{\mathcal{A}}(\langle\langle E, < \rangle, \phi \rangle)$ given in [Definition 19](#).

Monotonicity of ψ w.r.t. $SPOC_{\xi}(r)$ is obtained by direct application of [Lemma 5](#). ψ is injective by definition of $SPOC_{\xi}(r)$. The second requirement of [Definition 19](#) is that ψ must satisfy, for any $(t_1, t_2) \in dom(\phi)$ with $t_1 = v_1 \xrightarrow{e_1} v'_1$ and $t_2 = v_2 \xrightarrow{e_2} v'_2$:

$$e_{\psi(t_1)+1} \dots e_{\psi(t_2)-1} \in L^F(\mathcal{A}_{v'_1} \parallel \phi(t_1, t_2)).$$

With respect to $SPOC_{\xi}(r)$, we apply [Lemma 7](#) and obtain

$$\rho^F([\mathcal{A} \parallel \mathcal{O}_P]_{NCSr(t_1, t_2)}) = \rho^F(\mathcal{A}_{v'_1} \parallel \phi(t_1, t_2)). \quad (10)$$

The run r' has more effective choice transitions than r (9). We prove that there are fewer runs that preserve the effective choice transitions in $pre(t_1)$ of $SPOC_{\xi}(r')$ than runs that preserve the effective choice transitions in $pre(t_1)$ of $SPOC_{\xi}(r)$.

The upper bounds $\bar{\xi}_1^{(r')}$ are more restrictive than $\bar{\xi}_1^{(r)}$ for the same effective choice transition, formally (9) implies $\bar{\xi}_1^{(r')}(t_1) \leq \bar{\xi}_1^{(r)}(t_1) \wedge \bar{\xi}_2^{(r')}(t_1) \leq \bar{\xi}_2^{(r)}(t_1)$ by [Definition 15](#) of $\bar{\xi}_1^{(r')}$, $\bar{\xi}_2^{(r')}$, $\bar{\xi}_1^{(r)}$ and $\bar{\xi}_2^{(r)}$. Therefore, we have $NCSr'(t_1, t_2) \subseteq NCSr(t_1, t_2)$ ([Definition 16](#)). The runs that remain in those states are also included:

$$\rho^F([\mathcal{A} \parallel \mathcal{O}_P]_{NCSr'(t_1, t_2)}) \subseteq \rho^F([\mathcal{A} \parallel \mathcal{O}_P]_{NCSr(t_1, t_2)}). \quad (11)$$

One of those runs is

$$v_{\psi(t_1)} \xrightarrow{e_{\psi(t_1)+1}} v_{\psi(t_1)+1} \dots \xrightarrow{e_{\psi(t_2)-1}} v_{\psi(t_2)-1} \in \rho^F([\mathcal{A} \parallel \mathcal{O}_P]_{NCSr'(t_1, t_2)}),$$

therefore by (11),

$$v_{\psi(t_1)} \xrightarrow{e_{\psi(t_1)+1}} v_{\psi(t_1)+1} \dots \xrightarrow{e_{\psi(t_2)-1}} v_{\psi(t_2)-1} \in \rho^F([\mathcal{A} \parallel \mathcal{O}_P]_{NCSr(t_1, t_2)}).$$

We conclude with the application of [Lemma 7](#) that makes the correspondence between $[\mathcal{A} \parallel \mathcal{O}_P]_{NCSr(t_1, t_2)}$ and $\mathcal{A}_{v'_1} \parallel \phi(t_1, t_2)$ to obtain

$$e_{\psi(t_1)+1} \dots e_{\psi(t_2)-1} \in L^F(\mathcal{A}_{v'_1} \parallel \phi(t_1, t_2))$$

The last requirement in [Definition 19](#) is therefore satisfied, and we have $tr' \in sem_{\mathcal{A}}(CE(\mathcal{A}, \mathcal{O}_P, v_0, tr_1))$. $\square$

The following theorem proves the soundness of choice explanations as defined by Property 1 with respect to $\mathcal{O}_P$.

Theorem 5 (Soundness of choice explanations). *Given a behavioural model $\mathcal{A}$, an observer $\mathcal{O}_P$ of the safety property P , for any run $r = v_0 \xrightarrow{e_1} \dots \xrightarrow{e_n} v_n \in \rho^F(\mathcal{A} \parallel \mathcal{O}_P, v_0)$, the choice explanation $CE(\mathcal{A}, \mathcal{O}_P, v_0, tr(r))$ is sound with respect to $\mathcal{O}_P$.*

Given a choice explanation, soundness is implied by the existence of a maximal effective choice transition in $SPOC_\xi(r)$ that occurs in any run whose trace belongs to the semantics of the explanation and that leads to a state with level of choice $\langle 0, 0 \rangle$. In such state the violation is inevitable.

Proof. Let $tr = tr(r)$ and $CE(\mathcal{A}, \mathcal{O}_P, v_0, tr)$ be its choice explanation (Definition 18). Let ξ be the level of choice of $[\mathcal{A} \parallel \mathcal{O}_P]$ and $r' = v_0 \xrightarrow{e_1} v_1 \xrightarrow{e_2} \dots \xrightarrow{e_n} v_n$ be a run in $\rho(\mathcal{A} \parallel \mathcal{O}_P, v_0)$ such that $tr(r') \in sem_{\mathcal{A}}(CE(\mathcal{A}, \mathcal{O}_P, v_0, tr))$. Let t be the unique maximal element of $SPOC_\xi(r)$ (Lemma 6). By Theorem 4, t is also an effective choice transition of r' . We have $t = v_{i-1} \xrightarrow{e_i} v_i$ with $\xi(v_i) = \langle 0, 0 \rangle$ for some $i \leq n$. Therefore, with Lemma 4, the violation of P is inevitable from v_i . Hence, $tr(r') \notin safeLang_{\mathcal{A}, \emptyset}(\mathcal{O}_P, v_0)$. $\square$

The next theorem establishes completeness (Property 2): the explained trace is always contained in the semantics of its choice explanation.

Theorem 6 (Completeness of choice explanations). *Given a behavioural model $\mathcal{A}$, an observer $\mathcal{O}_P$ of the safety property P , for any run $r = v_0 \xrightarrow{e_1} \dots \xrightarrow{e_n} v_n \in \rho^F(\mathcal{A} \parallel \mathcal{O}_P, v_0)$, we have for $tr = tr(r)$:*

$$tr \in sem_{\mathcal{A}}(CE(\mathcal{A}, \mathcal{O}_P, v_0, tr)).$$

The proof is the direct application of Theorem 4 with $r' = r$.

Remark 1. As CE returns a unique explanation, rather than a set of explanations as in the general framework of Section 4, we reformulate the expected properties below for an explanation function returning a single explanation.

Theorem 7 (Precision of choice explanations). *Consider two observers $\mathcal{O}_{P_1}, \mathcal{O}_{P_2}$ with disjoint sets of private variables. Choice explanations are explanation-precise with respect to $\mathcal{O}_{P_1}$ and $\mathcal{O}_{P_2}$ and semantics-precise.*

Proof. Consider a model $\mathcal{A}$ and traces tr_1 and tr_2 of $\mathcal{A}$.

Semantics-precision: we have to show that for $\varepsilon_i = CE(\mathcal{A}, \mathcal{O}_{P_i}, v_{0,i}, tr_i)$, $i = 1, 2$,

$$L^F(\mathcal{A} \parallel \mathcal{O}_{P_1} \parallel \mathcal{O}_{P_2}, v_0) = \emptyset \implies sem_{\mathcal{A}}(\varepsilon_1) \cap sem_{\mathcal{A}}(\varepsilon_2) = \emptyset$$

By Theorem 4, the run of any trace of the semantics of ε_i reaches — via the effective choice transition reaching level $\langle 0, 0 \rangle$ — a state from which the accepting state of $\mathcal{A} \parallel \mathcal{O}_{P_i}$ is inevitable. As the accepting languages of $\mathcal{A} \parallel \mathcal{O}_{P_1}$ and $\mathcal{A} \parallel \mathcal{O}_{P_2}$ are disjoint, the claim follows.

Explanation-precision follows directly from semantics-precision of choice explanations. $\square$

The next theorem states the monotony of choice explanations defined in Property 4.

Theorem 8 (Monotony of choice explanations). *Given a behavioural model $\mathcal{A}$, an observer $\mathcal{O}_P$ of the safety property P , for any $tr_1, tr_2 \in L^F(\mathcal{A} \parallel \mathcal{O}_P)$,*

$$\begin{aligned} tr_2 \in sem_{\mathcal{A}}(CE(\mathcal{A}, \mathcal{O}_P, v_0, tr_1)) &\implies \\ sem_{\mathcal{A}}(CE(\mathcal{A}, \mathcal{O}_P, v_0, tr_2)) &\subseteq sem_{\mathcal{A}}(CE(\mathcal{A}, \mathcal{O}_P, v_0, tr_1)) \end{aligned}$$

The proof is obtained by applying Theorem 4 twice.

Proof. Let $tr_1, tr_2 \in L^F(\mathcal{A} \parallel \mathcal{O}_P, v_0)$, $tr \in L(\mathcal{A} \parallel \mathcal{O}_P, v_0)$ and r_1, r_2, r be the respective runs of tr_1, tr_2, tr in $\mathcal{A} \parallel \mathcal{O}_P$, such that $tr_2 \in sem_{\mathcal{A}}(CE(\mathcal{A}, \mathcal{O}_P, v_0, tr_1)) \wedge tr \in sem_{\mathcal{A}}(CE(\mathcal{A}, \mathcal{O}_P, v_0, tr_2))$.

By Theorem 4,

$$\begin{aligned} ECT_{\xi_1}(r_1) &\subseteq ECT_{\xi_1}(r_2) \wedge ECT_{\xi_2}(r_1) \subseteq ECT_{\xi_2}(r_2), \\ ECT_{\xi_1}(r_2) &\subseteq ECT_{\xi_1}(r) \wedge ECT_{\xi_2}(r_2) \subseteq ECT_{\xi_2}(r), \end{aligned} \tag{12}$$

hence

$$ECT_{\xi_1}(r_1) \subseteq ECT_{\xi_1}(r) \wedge ECT_{\xi_2}(r_1) \subseteq ECT_{\xi_2}(r). \tag{13}$$

Applying Theorem 4 again on the last equation, we have

$$tr \in sem_{\mathcal{A}}(CE(\mathcal{A}, \mathcal{O}_P, v_0, tr_1)).$$

We have hence proved that

$$sem_{\mathcal{A}}(CE(\mathcal{A}, \mathcal{O}_P, v_0, tr_2)) \subseteq sem_{\mathcal{A}}(CE(\mathcal{A}, \mathcal{O}_P, v_0, tr_1)).$$

$\square$

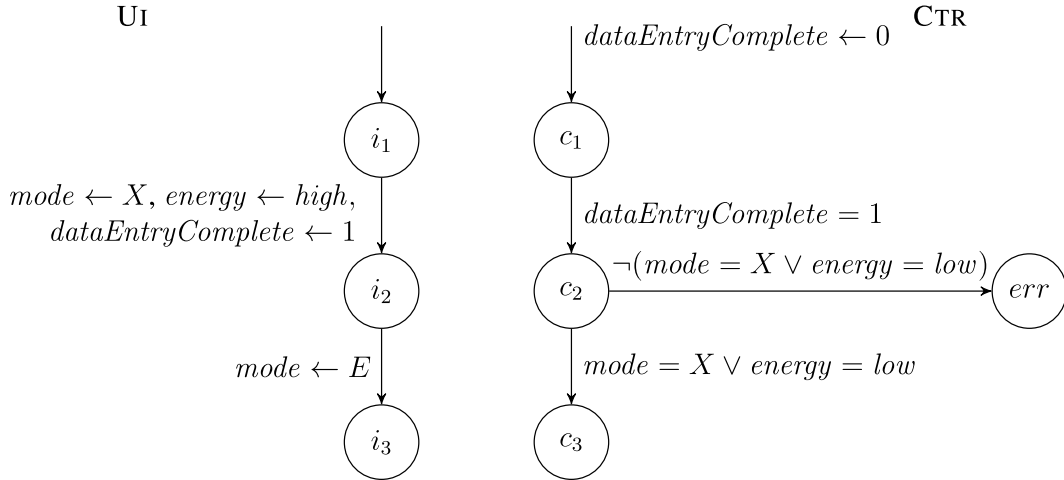


Fig. 11. Simplified model of the Therac-25 control software. The expected safety property is $(c = c_3 \implies \text{mode} = X \vee \text{energy} = \text{low})$.

5.8. Complexity

Computing a choice explanation as defined in Definition 18 requires computing the level of choice ξ , $SPOC_\xi$, and the non-choice automata ϕ .

We compute both level of choice functions ξ_i , $i = 1, 2$, independently using Algorithm 1. For a semantic graph $G = (V, \rightarrow) := [\mathcal{A} \parallel \mathcal{O}_P]$, the cost of computing ξ is in $O(|V|^2)$, since the body of the **while** loop is executed once for each node, with a cost of $O(|V|)$ for each execution. In order to avoid exploring the successors of state v in line 11, the current levels of choice χ_i of the successors can be kept in a multiset implemented as a hash table. As ξ only depends on the model and the safety property, but not on any run, it can be pre-computed once.

In contrast, the data structures $SPOC_\xi$ and ϕ are specific to each run to be explained. Since possible cycles in a run loop back to the level of choice of a state that has been visited before, they do not add any pertinent information to the choice explanation. When computing the choice explanation, we can therefore restrict ourselves to the acyclic part of the run. Let r be an acyclic (part of a) run. Hence, $|r| \leq |V|$.

Lemma 8. *The width of $SPOC_\xi(r)$, i.e. $\max\{|E| \mid E \subset ECT_\xi(r) \wedge \forall (t, t') \in E, \neg(t < t')\}$, is at most 2.*

Proof. Let us take three different transitions in E . For some $i \in \{1, 2\}$, two of those transitions t, t' are contained in $ECT_{\xi_i}(r)$ and therefore $t < t' \vee t' < t$. The claim follows. $\square$

We have that $|ECT_\xi(r)| \leq |r|$. As the order relation $<$ of Definition 14 is acyclic, its transitive closure, and hence $SPOC_\xi(r)$, is computed in $O(|r|^2)$. By Lemma 8, there are $O(|r|)$ non-choice implicit automata $\phi(t, t')$ in the explanation that can be computed jointly in $O(|V| \times |r|)$.

Hence, the overall complexity of the computation specific to run r is in $O(|r|^2 + |V| \times |r|) = O(|V| \times |r|)$.

6. A second case study

Multi-threaded concurrent programs are notoriously hard to debug since the outcome may depend on the scheduling of threads. For instance, one of the well-known *Therac-25* failures (see e.g. [20]) led to injury or death of several patients. A simplified model of Therac-25 is shown in Fig. 11: the operator inputs treatment data to the user interface (left automaton UI) and then changes the mode from X-ray to electron beam. The control part (right automaton CTR) waits for an input to be completed and runs a safety check. If successful, treatment is started (state c_3), otherwise an error state is entered. All events are controllable and observable.

This system can be represented by the following implicit automata:

$$\begin{aligned} \text{UI} = & \langle \{data, emode\}, \{mode, energy, dataEntryComplete, i\}, true, \delta_{\text{UI}}, \\ & v_{\text{UI}}^0 = (mode = X, energy = low, dataEntryComplete = 0, i = i_1), true \rangle \end{aligned}$$

with

- $\delta_{\text{UI}}(data) = (i = i_1, (mode \leftarrow X, energy \leftarrow high, dataEntryComplete \leftarrow 1, i \leftarrow i_2))$
- $\delta_{\text{UI}}(emode) = (i = i_2, (mode \leftarrow E, i \leftarrow i_3))$.

$$\text{CTR} = \langle \{chk_data, error, start\}, \{mode, energy, dataEntryComplete, c\}, true,$$

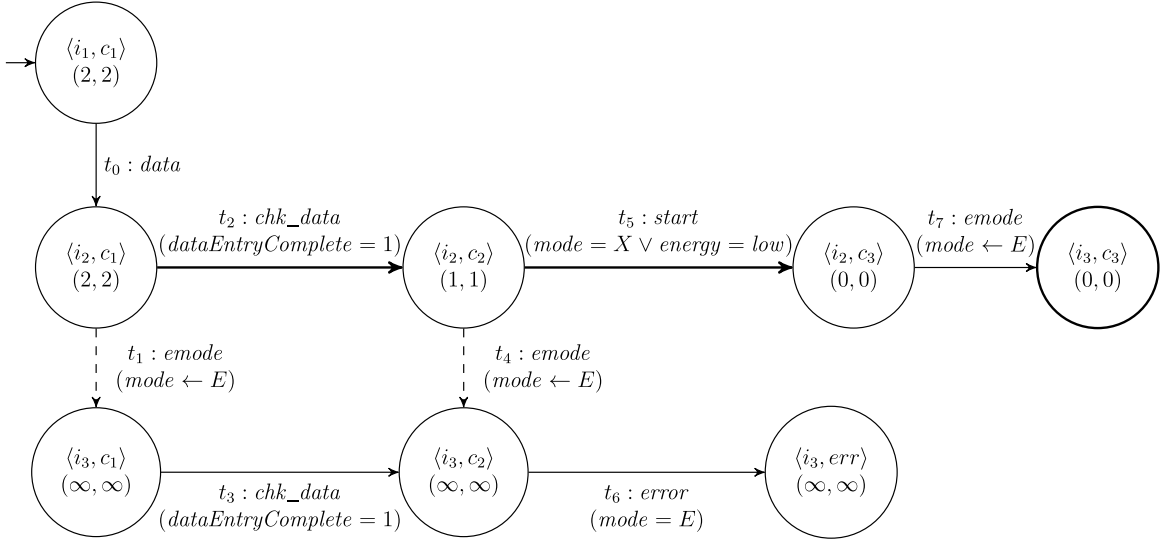


Fig. 12. Semantic graph of the therac model associated with the level of choices (ξ_1, ξ_2) (printed in the states). Thick states represent states where the safety property is violated. Effective choice transitions are printed with thick lines and safe alternatives are printed with dashed lines.

$$\delta_{\text{CTR}, \nu_{\text{CTR}}^0} = (\text{mode} = X, \text{energy} = \text{low}, \text{dataEntryComplete} = 0, c = c_1, \text{true})$$

with

- $\delta_{\text{CTR}}(\text{chk_data}) = (c = c_1 \wedge \text{dataEntryComplete} = 1, (c \leftarrow c_2))$
- $\delta_{\text{CTR}}(\text{error}) = (c = c_2 \wedge \neg(\text{mode} = X \vee \text{energy} = \text{low}), (c \leftarrow \text{err}))$
- $\delta_{\text{CTR}}(\text{start}) = (c = c_2 \wedge (\text{mode} = X \vee \text{energy} = \text{low}), (c \leftarrow c_3))$

The expected safety property ensures that, when the controller starts the treatment, the mode is set to X (X-ray), or the energy rate is low. This is formally defined by:

$$c = c_3 \implies (\text{mode} = X \vee \text{energy} = \text{low}).$$

Here, the proposed observer $\mathcal{O}_P$ for this property is defined as follows:

$$\mathcal{O}_P = \{\{\text{emode}\}, \{\text{pko}, \text{energy}, c\}, \text{true}, \delta_{\mathcal{O}_P}, (\text{energy} = \text{low}, c = c_1, \text{pko} = \text{false}), \text{pko}\}$$

with

- $\delta_{\mathcal{O}_P}(\text{emode}) = (\text{true}, (\text{pko} \leftarrow \text{true}, \text{ if } \text{energy} = \text{high} \wedge c = c_3)).$

Fig. 12 shows the semantic graph $[A \parallel \mathcal{O}_P]$ annotated with the levels of choice. In this simplified model, only one state violates the safety property (represented with a thick circle).

Considering now that the current scenario that is observed is actually the run that violates the safety property: $r = t_0 t_2 t_5 t_7$. This run is associated with the trace $tr = \text{data}, \text{chk_data}, \text{start}, \text{emode}$. The choice explanation of this run is $CE(\text{UI} \parallel \text{CTR}, \mathcal{O}_P, \nu_{\text{UI}}^0 \cup \nu_{\text{CTR}}^0, tr) = \langle \text{SPOC}_{\xi}(r), \phi \rangle$ where:

- $\text{SPOC}_{\xi}(r) = \{t_2 < t_5\};$
- Let ν be the target state of t_2 , it is also the source state of t_5 and $\nu = (\text{mode} = X \wedge \text{dataEntryComplete} = 1 \wedge c = c_2 \wedge i = i_2 \wedge \text{pko} = \text{false})$. Moreover $\text{NCS}^r(t_2, t_5) = \{\nu\}$. It follows that

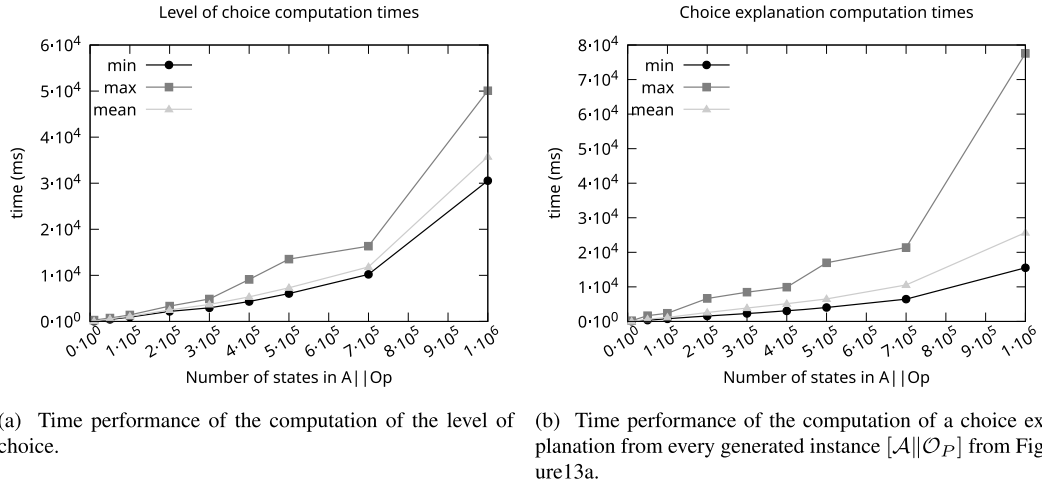
$$\begin{aligned} \phi(\langle t_2, t_5 \rangle) &= \langle \{\text{emode}\}, \{\text{mode}, \text{dataEntryComplete}, c, i, \text{pko}, \text{energy}\}, \\ &\quad \text{inv} = \nu, \delta_{\mathcal{O}_P}, F = (\text{mode} = X \wedge \text{dataEntryComplete} = 1 \\ &\quad \wedge c = c_2 \wedge i = i_2) \rangle \end{aligned}$$

Informally speaking, this choice explanation states that the safety property has been violated by the controller that has consecutively checked the input data (t_2) and started the treatment (t_5) while the user has not yet set the mode E . The action t_1 by the user was a safe alternative to the effective transition choice t_2 while the action t_4 was a safe alternative to the effective transition choice t_5 (see the dashed transitions in Fig. 12).

Table 4

Sizes of the semantic graphs of the behaviour model $\mathcal{A}$ and its different compositions with observers $\mathcal{O}_P$, $\mathcal{O}_Q$ and the log $\mathcal{L}$ involved in the running example from Section 3.

	$[\mathcal{A}]$	$[\mathcal{A} \parallel \mathcal{O}_P]$	$[\mathcal{A} \parallel \mathcal{O}_P \parallel \mathcal{L}]$	$[\mathcal{A} \parallel \mathcal{O}_P \parallel \mathcal{O}_Q]$	$[\mathcal{A} \parallel \mathcal{O}_P \parallel \mathcal{O}_Q \parallel \mathcal{L}]$
States	676	676	31	1291	31
Transitions	1372	1372	30	2646	30

**Fig. 13.** Overall performance.

7. Experimental evaluation

The full set of algorithms for the construction of choice explanations has been implemented in a C++ prototype that is available as open-source in the gitlab repository <https://gitlab.laas.fr/ypencole/ddexplain>. All the examples described in this paper have been designed and tested with this implementation. Table 4 shows the sizes of the semantic graphs involved in the two-car problem introduced in Section 3.

This section reports on some experimental results based on the random generation of semantic graphs $[\mathcal{A} \parallel \mathcal{O}_P]$ in order to evaluate the practical performance of the pre-computation of the levels of choice on one hand and the computation of choice explanations for a given run on the other. The gitlab repository contains all the programs to reproduce these experiments. The results presented in this section are obtained on a Intel Core i9-12900K (16 cores) 128Go 3600MHz DDR5 running the Debian 13 operating system.

Fig. 13a shows the time performance with a polynomial trend of the computation of the level of choice with respect to the number of states n in $[\mathcal{A} \parallel \mathcal{O}_P]$, $10000 \leq n \leq 1000000$. For each size n , 200 instances of $[\mathcal{A} \parallel \mathcal{O}_P]$ have been randomly generated. Fig. 13a presents the minimal/maximal/mean time for computing their respective level of choice. In each generated instance, 100% of the events are observable, 75% of the events are controllable ($cRatio = 0.75$). The number of output transitions of each non-violating state is randomly selected between 1 and $bMax = 10$ ensuring that a non-violating state always has a successor. On the other hand, the violating states have been randomly selected with a ratio of $vRatio = 0.0001$ of the n states (1 for $n = 10000$, 10 for $n = 100000$...) and the number of output transitions of these violating states is set between 0 and $bMax = 10$. The random generation ensures that a violating state is either a deadlock or any of its successors is a violating state. It follows that the number of transitions in $[\mathcal{A} \parallel \mathcal{O}_P]$ is around $5n$: for instance, the greatest number of transitions in the $[\mathcal{A} \parallel \mathcal{O}_P]$ of size $n = 1000000$ generated to produce Fig. 13a is 5546374.

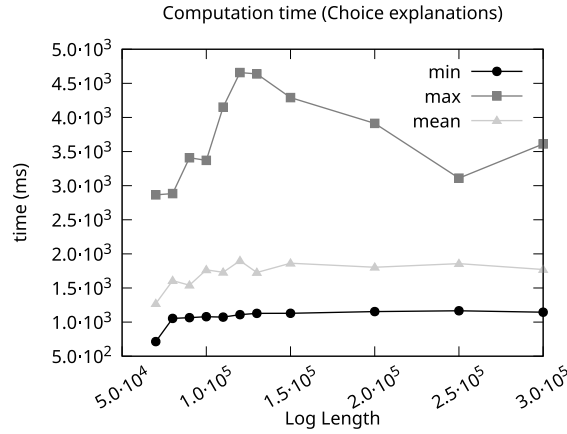
For the sake of illustration, Fig. 13b shows for each generated instance from Fig. 13a, the computation time of a choice explanation for a selected violating run of the instance. The selected run has been selected randomly and leads to one final state that violates the property. To ensure the selected run is realistic with respect to the system, the length of the run has been set to be roughly twice the number of transitions of the generated $[\mathcal{A} \parallel \mathcal{O}_P]$ to ensure the run does not only simulate transient but also permanent behaviours (in particular, the selected run necessarily involves cycles as its length is greater than the number of states in $[\mathcal{A} \parallel \mathcal{O}_P]$ by construction): for instance, the run generated from $[\mathcal{A} \parallel \mathcal{O}_P]$ with the greatest number of transitions is 11092759. The maximal time for computing choice explanations in these illustrative examples is below 80 seconds.

In order to measure the number of levels of choice and the size of the choice explanations with respect to the size of $[\mathcal{A} \parallel \mathcal{O}_P]$ and its level of controlability, another set of experiments based on random $[\mathcal{A} \parallel \mathcal{O}_P]$ and run generations have been conducted. Each generated $[\mathcal{A} \parallel \mathcal{O}_P]$ is based on a set of parameters.

Table 5

Range of maximal levels of choice ($\lfloor \xi_i^{max} \rfloor \leq \xi_i^{max} \leq \lceil \xi_i^{max} \rceil$) $_{i \in \{1,2\}}$, range of sizes of choice explanations (number of effective choice transitions: $\lfloor ECT \rfloor \leq ECT \leq \lceil ECT \rceil$) and size means $m(ECT)$.

$ \mathcal{A} \mathcal{O}_P $	$\lfloor \xi_1^{max} \rfloor$	$\lceil \xi_1^{max} \rceil$	$\lfloor \xi_2^{max} \rfloor$	$\lceil \xi_2^{max} \rceil$	$\lfloor ECT \rfloor$	$\lceil ECT \rceil$	$m(ECT)$
100	0	10	2	10	1	11	2
200	0	8	2	9	1	11	2
500	0	7	2	7	1	9	2
1000	0	6	2	7	1	8	2

**Fig. 14.** Explanation computation time wrt log length, (size of $|\mathcal{A}||\mathcal{O}_P|$: 100000 states).

- $size \in \{100, 200, 500, 1000\}$ is the number of states in $|\mathcal{A}||\mathcal{O}_P|$.
- $bMax \in \{3, 20\}$ is the maximal number of output transitions per state in $|\mathcal{A}||\mathcal{O}_P|$. For each state of a generated $|\mathcal{A}||\mathcal{O}_P|$, the number of output transitions is randomly selected within $[1, bMax]$.
- $cRatio \in \{0.25, 0.5, 0.75, 0.95, 1\}$ is the ratio between the number of controllable events and the total number of events in $|\mathcal{A}||\mathcal{O}_P|$.
- $vRatio \in \{0.1, 0.01\}$ is the ratio between the number of states that violate the safety property and the total number of states.
- The set of events are observable.

These parameters and their ranges lead to the definition of 80 tested configurations $size \times bMax \times cRatio \times vRatio$. For each configuration, 100 $|\mathcal{A}||\mathcal{O}_P|$ have been generated (i.e. 8000 generated $|\mathcal{A}||\mathcal{O}_P|$) and for each generated $|\mathcal{A}||\mathcal{O}_P|$, 100 choice explanations have been computed based on random run simulations whose run length is roughly twice the number of transitions in the generated $|\mathcal{A}||\mathcal{O}_P|$ (i.e. 800,000 run simulations and explanation computations). Table 5 summarises the range of maximal levels of choice $\xi_i^{max}, i \in \{1, 2\}$ and the set of explanation sizes (as the number of effective choice transitions in their SPOC) that are obtained. Value ξ_i^{max} is the greatest value for the level of choice ξ_i in a specific $|\mathcal{A}||\mathcal{O}_P|$ and $\lfloor \xi_i^{max} \rfloor, \lceil \xi_i^{max} \rceil$ denote the minimal/maximal bounds of this value obtained after the generation of 8000 $|\mathcal{A}||\mathcal{O}_P|$. In these experiments, it must be noticed that the maximal levels of choice $\langle \xi_1^{max}, \xi_2^{max} \rangle$ were never greater than 10. $|\mathcal{A}||\mathcal{O}_P|$ models with higher maximal levels of choice are possible but are unlikely. As expected, Table 5 illustrates that the size of the choice explanations (i.e. the number of effective choice transitions ECT) is always lower or equal to the sum of $\xi_1^{max} + \xi_2^{max}$, hence the concision of the explanation with respect to the explained run. It shows that the size of the choice explanations does not directly depend on the size of $|\mathcal{A}||\mathcal{O}_P|$ and the length of the run it explains but on the maximal levels of choice that is likely rather low.

Finally, Fig. 14 depicts an experiment that shows the computation time of an explanation with respect to the length of the run that is explained. In this figure, one random $|\mathcal{A}||\mathcal{O}_P|$ composed of 100000 states/554077 transitions has been generated with parameters $bMax = 10$, $cRatio = 0.75$, $vRatio = 0.0001$, similarly as in Fig. 13a. For each selected log length, 100 random simulations have been performed. Fig. 14 shows that ultimately, the computation of the explanation does not depend on the length of the run. This is because cycles in the run do not contain any effective choice transitions. The latter are only present in the acyclic parts of the run, whose size is bounded by the number of states in $|\mathcal{A}||\mathcal{O}_P|$. This information is encoded by the precomputed levels of choice over $|\mathcal{A}||\mathcal{O}_P|$.

As a conclusion, these experiments show on one hand that the computation of the levels of choice scales up with respect to the size of semantic graphs as well as the computation of choice explanations. Results also show how concise an explanation can be as its size only depends on the maximal level of choices in $|\mathcal{A}||\mathcal{O}_P|$, and not on the length of the log $\mathcal{L}$ to explain.

8. Related work

This work is related to the fault diagnosis problem in DES as originally introduced in [15]. In this seminal study, the system is modeled as an automaton explicitly representing both normal and abnormal behaviours. Specifically, a system run is considered abnormal if it contains an unobservable fault event. The fault diagnosis problem then involves identifying the set of fault events that could have occurred in the partially observable runs consistent with a given log L . In the original fault diagnosis problem, there is no attempt to provide any kind of detailed explanation of why the fault happened, the only relationship between the diagnosed fault and the observations is the consistency with the model and the log. Some approaches reconstruct the complete set of faulty runs consistent with the observations [16,22] without attempting to synthesise more concise explanations. Other approaches retain only one preferred run out of the set of candidates [23].

Fault diagnosis has also been extended to address complex behaviours, similar to our safety violations, referred to as *logic specifications* in [24] or *event patterns* in [25,26]. However, the results in these approaches are limited to identifying the set of possible runs consistent with the observations. In contrast, the work in [27] focuses on providing interpretable explanations for diagnosed faults by identifying a subset of observations called *critical observations*. Similar to choice explanations, critical observations can be represented as an automaton but are restricted to observable events. Critical observations form a sufficient condition for the presence of the fault F . Critical observations are merely correlated with the fault. The present work adopts a similar objective, aiming to provide interpretable explanations for safety violations in partially observable systems. However, our approach is causal, and it extends beyond fully observable explanations. We propose causal explanations that characterise all system runs leading to the safety violation, including unobserved events. This approach offers comprehensive, yet concise, interpretable explanations that highlight both observed and inferred events, thereby shedding light on why an undesirable state occurred — even if the causes were not directly observed.

Our proposal for generating explanations is also related to the *reveals* relation in a DES as defined in [28]. An event e reveals an event e' if e' occurs in any run of the system where e occurs: the occurrence of e is then a sufficient condition for the occurrence of e' . More generally speaking, explanation is a notion that is intrinsic to abductive diagnosis problems as logically defined in [29,30] and more recently in [31]. In an abductive diagnosis problem, the purpose is to explain a set of *symptomatic observations* $OBS_S \subseteq L$ by fault modes of the system usually represented by abducible predicates. Here the relationship between the explanation and the observation is logical entailment ($\models$) that is stronger than simple consistency. This means that there might be no diagnosis candidate (so no explanation) for a given OBS_S . Our choice explanations are inherently abductive, following the logic of abductive diagnosis, where the symptomatic observation is just the violation of a partially observable property.

Several authors have proposed a construction of explanations for the satisfaction of a reachability property by an execution trace based on subsequences of the trace that are sufficient to entail the property, such as explanatory diagnoses in the Situation Calculus [1] and causal compression in rule-based rewriting systems [2]. Subsequence explanations are also used in Explainable Reinforcement Learning (XRL) [32] as parts of executions of an AI agent that comply with some specifications on the initial and final states. A nondeterministic finite automaton is built on the query about the agent's behaviour and the input traces are tested on it to find subsets accepted by the automaton. However, subsequence explanations do not convey any information about the outcome produced by alternative behaviours of the system. They also exhibit numerous limitations concerning the desired properties of explanations. Specifically, they cannot be both complete and precise, and the relevance of certain events in the explanation is open to question. These issues are addressed by choice explanations.

Our definition of effective choices is inspired by [3], which uses game theory to explain counterexample traces from model checking. This approach highlights portions of the trace where the system could have avoided violating an expected safety property, regardless of the environment's behaviour. However, our approach disregards events that are “compensated” by other events later in the trace, hence yielding more concise explanations. A similar idea of highlighting choice states in which the run could have taken a different outcome is followed in [33], but without the fixpoint construction of our approach, also found in [3], transitions that contribute to the system failure but that alone are not fatal, are discarded from the explanation there. Also worth noting is that [3,33] assume full observability in contrast to our approach.

More recently, [34] has proposed an approach based on Petri net unfoldings to identify “cliff edges” in configurations, similar to a level of choice of 1 in our approach. *Protectedness* is then defined as the number of decisions separating the current state from falling off the cliff. In contrast to our approach, protectedness does not take into account the controllability of transitions.

Our formalism of *SPOC*, used to symbolically represent an explanation as a set of *executions*, has similarities with *Event Order Logic*, introduced in [35] to represent a set of counterexample *traces* (in the sense of our Definition 4) as partially ordered sets of events. *SPOC*, in contrast, are ordered sets of transitions. This finer granularity is required for our explanations to satisfy the expected properties, including soundness (Theorem 5) and completeness (Theorem 6).

Choice explanations are based on *counterfactual* causality: in order to compute the level of choice of a branching state, and determine causality of the choice made in the actual world, the futures in the alternative branches have to be taken into account. In this point our construction of counterfactuals differs from definitions such as Halpern and Pearl's actual causality [14] and their adaptations to temporal causes such as [36]: in actual causality, counterfactual analysis relies on “mini-surgeries” in the form of interventions that change the behaviour from the (fully deterministic) actual model to a different model. In contrast, our goal is to explain failures in terms of branching choices, hence, interventions merely amount to choosing a different branch within the same model.

[37] gives a definition of temporal causality in reactive systems that is inspired by [14]. In a similar spirit, Sallinger et al. [38] defines counterfactual causes of “Heisenbugs” as a minimal subset of inputs that determine whether a safety property is violated or

not. Again, those works do not explain *how* a set of causes entails the violation of an expected property. Parreaux et al. [39] defines counterfactual causation and causal explanation in two-player games based on a distance metric, in line with Lewis' seminal work [12]. This definition encompasses a quantification over a set of closest strategies, which is expensive to compute.

Baier et al. [40] uses generalised power indices to attribute responsibility for the violation of a safety property to the states of an automaton. While there are similarities to our work — the goal of both approaches is to establish a form of *backward responsibility* van de Poel [41] for the outcome —, the two approaches differ fundamentally in several aspects. The goal of [40] is to quantify the responsibility of states, whereas our objective is to identify the transitions that have caused the outcome. In addition, Baier et al. [40] counts the number of coalitions of states needed to change the outcome, with exponential complexity in the number of states, whereas the cost of our approach is quadratic.

Since the approach we develop in this paper is in the field of AI and provides explanations, one naturally makes the connection to the area of eXplainable Artificial Intelligence (XAI) [6,7]. The question of explaining automated decision-making systems is core both in symbolic and subsymbolic AI, with an intensive focus on the latter in the recent years. Some explanation properties overlap the two AI tracks, for instance causality. Parts of the subsymbolic XAI literature are indeed beginning to address causality — especially when black-box predictions concern action guidance [42], or planning as in eXplainable AI Planning (XAIP) [43,44].

This work does not address the explanation of black-box models such as neural networks [7] for which associational measures, such as feature attribution methods [45], aim at fostering trust in their predictions. Instead, we have complete knowledge of a white-box model that represents explicitly the alternative behaviours of a system. Our approach seeks to derive causal explanations from the model using its own interpretable components. Consequently, the techniques employed in our setting are fundamentally different in nature and purpose from those commonly used in XAI methods like Shap or Lime [46]. Nevertheless, some of the desirable properties that have been proposed for classifier explainers [47] share similarities with the properties that we propose for explanations. For instance, "coherence" for a class explainer, as defined in [48], conveys the same idea as "precision" in this paper. The two properties indicate that the explanations for two classes or for two property violations, respectively, have distinct explanations. Also distinguishing between controllable and uncontrollable events for choice explanations can possibly compare with mutable and immutable features for counterfactual methods [49]. These concepts are reflected in the feasibility property proposed, among others, by [50] for counterfactual explanations after classifying traces, a central concept in our approach. However, unlike the proposed properties, the ones we propose are formal and firmly grounded in the semantics of choice explanations. Referring to planning and XAIP, manipulated concepts such as states, actions, and transition functions align with the formalism employed in our approach. We believe that understanding a plan that achieves a specific goal state or explaining why certain actions are not feasible in a given state are challenges that could benefit from our explanation framework and the concept of safe alternatives.

9. Conclusion

We have presented a theory of explanations for safety violations in discrete-event systems. Our framework can handle partial observability in dynamic systems represented by implicit automata. These automata enable the integration of variables alongside events while keeping discrete states implicit. This formalism is rich enough to represent non trivial systems. It makes it possible to express (i) sophisticated safety properties that combine constraints on the sequential configuration of events, states, and constraints on variables, (ii) rich explanations combining events and state invariants expressed using variables. As a first step, we have discussed explanations based on minimal subsequences of the traces of the log that entail a violation of the safety property, and discussed the properties and shortcomings of this type of explanations. We have then introduced choice explanations. This approach is based on a construction of layers similar to [3], in which the explanation is constructed from the choices that drive the system closer to, and ultimately result in, the violation of the safety property. Both approaches are complementary: while subsequence explanations are well suited to "condense" the execution trace in sequential portions of the model but are prone to keep non-pertinent parts such as initialisation sequences in the explanation, choice explanations highlight the "fateful" choices in an execution. Alternative events that would have helped avoid the outcome are also highlighted. Choice explanations are therefore able to explain failures stemming from possible choices, like in the case of concurrency bugs.

In this work, and unlike existing works, we have defined explanations in terms of behaviours over events and variables, using information from states such as valuations of variables (or abstractions thereof) to convey additional information in the explanations. Other key contributions of our work include the formalisation of explanation semantics and the definition of expected properties of explanations in a general setting, proved to be satisfied by choice explanations. An efficient algorithm has been presented to obtain choice explanations. Its implementation is made available in the GitLab repository <https://gitlab.laas.fr/ypencole/ddexplain>.

Interesting questions remain open and are regarded as potential avenues for future investigation. Notably, constructing explanations on abstractions and in an on-the-fly manner is of particular interest, with the aim of developing accountable embedded systems. It is also crucial to identify the necessary safeguards, such as monitoring and diagnostic mechanisms, to prevent safety violations by leveraging explanations of those violations. This highlights the inherent interconnection between diagnosis and explanations. Deciding which observation should be included in the log to obtain a reduced set of explanations for a given run is also an interesting problem.

CRedit authorship contribution statement

Gregor Gössler: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Conceptualization; **Thomas Mari:** Writing – original draft, Validation, Methodology, Investigation, Conceptualization; **Yannick Pencolé:**

Writing – original draft, Validation, Software, Methodology, Investigation; **Louise Travé-Massuyès**: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix A. Implicit automata composition

Definition 20 ($\oplus$, consistent updates). Given two functions $f_i \in \text{update}(X)$, $i = 1, 2$, let

$$\begin{aligned} f_1 \oplus f_2 &= \{x \mapsto v \mapsto \mathbf{x} \mid x \mapsto v \mapsto \cdot \in (f_1 \setminus f_2) \cup (f_2 \setminus f_1) \\ &\quad \vee (x \mapsto v \mapsto \mathbf{x}) \in f_1 \cap f_2\} \end{aligned}$$

be their composition.

f_1 and f_2 are *consistent* if $\forall x \in \text{dom}(f_1) \cap \text{dom}(f_2)$, $\text{dom}((f_1 \oplus f_2)(x)) = \text{dom}(f_1(x)) \cup \text{dom}(f_2(x))$.

Definition 21 (Composition of implicit automata). Let $\mathcal{A}_i = \langle \Sigma_i, X_i, \text{inv}_i, \delta_i, v_i^0, F_i \rangle$ for $i \in \{1, 2\}$, be two implicit automata and v_0 a state of $\mathcal{A}_1$ such that v_0 and v_2^0 are consistent, and for $\delta_i(e) = \langle g_i, f_i \rangle$, $i = 1, 2$ with $e \in \Sigma_1 \cap \Sigma_2$, f_1 and f_2 are consistent.

The composition of $\mathcal{A}_1$, initialised in v_0 , and $\mathcal{A}_2$ is the implicit automaton $\mathcal{A}_1 \parallel_{v_0} \mathcal{A}_2 = \langle \Sigma_1 \cup \Sigma_2, X_1 \cup X_2, \text{inv}_1 \wedge \text{inv}_2, \delta, v_0 \cup v_2^0, F_1 \times F_2 \rangle$ where

$$\begin{aligned} \delta &= \{ \langle e, \delta_1(e) \rangle \mid e \in \Sigma_1 \setminus \Sigma_2 \} \cup \\ &\quad \{ \langle e, \delta_2(e) \rangle \mid e \in \Sigma_2 \setminus \Sigma_1 \} \cup \\ &\quad \{ \langle e, \delta_1(e) \otimes \delta_2(e) \rangle \mid e \in \Sigma_1 \cap \Sigma_2 \} \end{aligned}$$

where $\langle g_1, f_1 \rangle \otimes \langle g_2, f_2 \rangle = \langle g_1 \wedge g_2, f_1 \oplus f_2 \rangle$. We omit the subscript $v_0 \parallel$ when $v_0 = v_1^0$.

In the above definition, events that only belong to one automaton $\mathcal{A}_i$ are considered as asynchronous in the composition, whereas events shared by both automata are synchronised. Therefore the transition relation δ of the composition consists of:

1. all the transitions of $\mathcal{A}_1$ and $\mathcal{A}_2$ labelled with events e that are not shared by both automata;
2. the transitions resulting from the $\otimes$ composition of a transition from $\mathcal{A}_1$ and a transition from $\mathcal{A}_2$ labelled with a shared event e .

The composition makes sure that a transition labeled with a shared event is enabled if and only if the event is enabled in both source transitions and the update functions agree on their assignments to all shared variables in the state from which the transitions are issued.

Definition 22 (Projection of a semantic state). Given a composed implicit automaton $\mathcal{A}_1 \parallel \mathcal{A}_2 = \langle \Sigma, X, \text{inv}, \delta, v^0, F \rangle$ with $[\mathcal{A}_1 \parallel \mathcal{A}_2] = \langle \mathcal{V}, \Sigma, \rightarrow, v^0, \mathcal{V}^F \rangle$ and $v = \langle v_1, v_2 \rangle \in \mathcal{V}$, we define the projection of v on $\mathcal{A}_i$, $i = 1, 2$, as $v_{\downarrow \mathcal{A}_i} := v_i$.

As composition $\mathcal{A}_1 \parallel \mathcal{A}_2$ is associative, it is straightforward to extend projection to products of more than two implicit automata.

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